



Nonlinear self-excited forces and aerodynamic damping associated with vortex-induced vibration and flutter of long span bridges

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ABSTRACT

This study presents a comprehensive investigation of nonlinear self-excited forces and negative aerodynamic damping associated with vertical and torsional vortex-induced vibrations (VIVs) and coupled flutter of bridges. Wind tunnel test of a spring-supported streamlined box deck section model is carried out to characterize vertical and torsional VIVs and flutter at different angles of attack. A Hilbert Transform-based approach is proposed to determine the amplitude-dependent aerodynamic damping and other response characters. The self-excited forces in terms of flutter derivatives are subsequently extracted as a polynomial functions of vibration amplitude. The flutter derivatives are validated by comparing predicted VIV and flutter amplitudes with wind tunnel data. The VIV and flutter amplitudes of a three dimensional (3D) suspension bridge are further predicted with consideration of spanwise variation of self-excited forces. The influence of partial spanwise correlation of self-excited forces on VIV is also examined. The 3D analysis leads to a significant lower VIV and flutter amplitudes as compared to 2D analysis. The study not only presents an effective analysis approach and also provides new insights that lead to improve understanding and quantification of nonlinear self-excited forces and their effects on bridge VIV and flutter.

1. Introduction

Vortex-induced vibration (VIV) has been one of the important issues for aerodynamic studies of long span bridges. At the vortex lock-in wind speed, the self-excited forces introduce amplitude-dependent negative aerodynamic damping to structural system. The VIV reaches a steady-state amplitude when the negative aerodynamic damping is identical to the structural damping that leads to zero system damping. VIV has been experienced on several long span bridges, such as the Trans-Tokyo Bay Crossing Bridge in Japan, Great Belt Bridge in Denmark and Xihoumen Bridge in China. Frequent VIV may cause fatigue or discomfort, which directly affects bridge serviceability.

Great research efforts have been devoted to establish vortex-induced force models based on wind tunnel studies (e.g., Iwan and Blevins, 1974; Landl, 1975; Tamura and Matsui, 1979; Vickery and Basu, 1983; Simiu and Scanlan, 1986; Ehsan and Scanlan, 1990; Larsen, 1995; Mashad and Jones, 2014). Ehsan and Scanlan (1990) introduced an approach to determine the empirical nonlinear force model parameters using either a 'decay-to-resonance' or a 'growth to resonance' free vibration data obtained in wind tunnel. Chen (2013) adopted a nonlinear aerodynamic

damping model as a polynomial function of time-varying velocity and/or displacement of vibration, which is equivalent to the amplitude-dependent damping model in the case of harmonic motion. The response statistics can be determined analytically by using the method of equivalent nonlinear equation where the influence of stochastic buffeting force was included (Chen, 2014). Zhu et al. (2013) proposed a nonlinear force model with higher-order terms of both vibration displacement and velocity, and introduced a model parameter identification approach via a nonlinear least-square fitting of the measured growth to resonance response time history. An effort to reduce the complexity of the force model was made in Zhu et al. 2017a,b. To predict the amplitude of VIV of a full scale bridge, it is also critical to account for the spanwise correlation of vortex-induced forces for which more research efforts are needed (e.g., Wilkinson, 1981; Ehsan and Scanlan, 1990; Sun et al., 2013; Li et al., 2016; Zhu et al., 2017a,b; Zhou et al., 2017). Zhou et al. (2017) compared the VIV amplitude of a novel aeroelastic model with that of section model, and discussed the VIVs associated with higher vibration modes. More studies are needed on the modeling and identification of nonlinear self-excited forces and amplitude-dependent aerodynamic damping associated with VIV of

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bridge sections, and prediction of three dimensional (3D) bridge VIV with consideration 3D effects.

Flutter instability is another major concern in the design and construction of long span bridges that is excited by self-excited forces at relatively strong winds and can result in destruction of the structure. The self-excited forces are often modeled as linear functions of vibration displacement and velocity in terms of flutter derivatives which are independent of vibration amplitude. Great research efforts have been made on the quantification of flutter derivatives from wind tunnel testing, and determination of flutter onset wind speed and coupled motions using multimode coupled analysis approaches (e.g., Scanlan, 1978; Matsumoto et al., 1997, 1999; Chen et al., 2000a, 2000b). According to this linear flutter theory, the bridge system becomes dynamically unstable beyond the critical flutter wind speed, where the negative aerodynamic damping caused by self-excited forces exceeds the structural damping.

On the other hand, several wind tunnel studies of bridge sections at smooth flow have revealed amplitude-dependent flutter vibration. This nonlinear flutter is typically observed in the cases of relatively bluff sections and streamlined sections with a large angle of attack. Several nonlinear self-excited force models have been proposed with the aim to predict the amplitude of nonlinear flutter. Scanlan (1997) pointed out that flutter derivatives may be slightly amplitude dependent. Noda et al. (2003) founded the dependence of flutter derivative A_2^* and H_2^* of thin rectangular cylinders on vibration amplitude. Náprstek et al. (2007) introduced a generalized Van der Pol with Duffing type of nonlinear aerodynamic force model for prismatic bodies. Xu et al. (2016) investigated the influence of vibration amplitude on flutter derivatives of a flat plate and two bluff deck sections by forced vibration simulation with CFD. Zhang et al. (2017) proposed a nonlinear aerodynamic force model for nonlinear torsional flutter, where aerodynamic stiffness and damping, i.e., flutter derivative A_2^* and A_3^* , were given as summation of constant and quadratic function of torsional displacement. Zhou et al. (2018) investigated the nonlinear oscillation behaviors of a twin-box bridge girder with a sophisticated nonlinear force model involving many model parameters. Wu et al. (2019) presented a wind tunnel study of a truss girder section that exhibits high degree of flutter nonlinearity, and proposed an approach of extracting nonlinear aerodynamic damping and flutter derivatives. Other types of nonlinear aerodynamic force model including the effect of turbulence have also been developed (Diana et al., 2010; Chen et al., 2001, 2003a; Wu and Kareem, 2011, 2013; Carassale et al., 2014). Continued research efforts toward developing physically more meaningful and computational effective nonlinear aerodynamic force model and flutter analysis framework are required.

This paper presents a comprehensive study of nonlinear characteristics of vertical and torsional VIVs and coupled flutter of a streamlined box bridge section at different angles of attack through wind tunnel test and analysis. The section model is spring-supported in two degrees of freedoms (2DOFs) in both vertical and torsional directions. The free vibration response histories from rest to steady-state amplitude at different wind speeds and angles of attack are measured. A Hilbert Transform based approach is proposed to determine the amplitude-dependent aerodynamic damping and other response characteristics. Nonlinear flutter derivatives are subsequently extracted as functions of wind speed and vibration amplitude associated with VIVs and coupled flutter. The flutter derivatives are used to predict VIV and flutter amplitudes and then compared with wind tunnel data, which validates the accuracy of the proposed approach of extracting nonlinear flutter derivatives. The VIV characteristics of a three dimensional (3D) suspension bridge with consideration of spanwise variation of vibration amplitude and spanwise correlation of aerodynamic forces are predicted with the obtained nonlinear flutter derivatives. The limit cycle vibration amplitude of 3D bridge flutter is also estimated using the nonlinear flutter derivatives and compared to the 2D estimation. The nonlinear flutter characteristics at different angles of attack are discussed including the contributions of different components of self-excited forces to system damping.

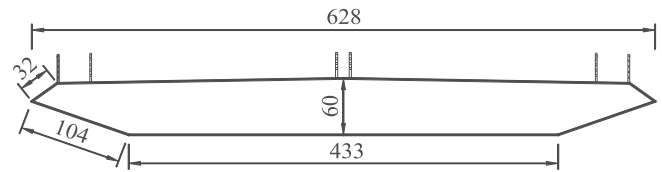


Fig. 1. Bridge deck section model (unit: mm).



Fig. 2. Model set-up in wind tunnel.

Table 1

Modal damping ratios of the bridge deck model.

Case 1		Case 2		Case 3		Case 4	
$\xi_{hs}(\%)$	$\xi_{as}(\%)$	$\xi_{hs}(\%)$	$\xi_{as}(\%)$	$\xi_{hs}(\%)$	$\xi_{as}(\%)$	$\xi_{hs}(\%)$	$\xi_{as}(\%)$
0.32	0.23	0.45	0.36	0.41	0.27	0.56	0.45
$\alpha = 0, 3, 5 \text{ deg}$		$\alpha = 0, 3 \text{ deg}$		$\alpha = 7 \text{ deg}$		$\alpha = 5, 7 \text{ deg}$	

2. Wind tunnel test of bridge section model

A streamlined bridge deck section model commonly used in long span suspension bridges was tested. Fig. 1 shows the details of the section model, which has a scale ratio of 1/50, $B = 2b = 0.628 \text{ m}$ in width, $D = 0.06 \text{ m}$ in height and $L = 2.1 \text{ m}$ in length. The wind tunnel test was carried out at the XNJD-2 wind tunnel of Southwest Jiaotong University, China. The test section is 2 m in height and 2.4 m in width. Fig. 2 shows the spring-supported system with 2DOFs in both vertical and torsional directions. The mechanical vibrations in vertical and torsional directions at zero wind speed are uncoupled. It is confirmed that the system remains linear with constant frequencies and damping ratios not affected by vibration amplitude in the interested range of magnitude of this study. The test was conducted under smooth flow using two sets of structural damping ratio at each angle of attack of $\alpha = 0, +3, +5$ and $+7^\circ$ (nose-up positive). The model test under a negative angle of attack was not implemented as there is no obvious VIV based on a previous study for this bridge section. The mass and mass moment of inertia per unit length of the model are $m = 6.524 \text{ kg/m}$ and $I = 0.214 \text{ kg.m}^2/\text{m}$. The vertical and torsional natural frequencies of the system are $f_{hs} = 1.294 \text{ Hz}$ and $f_{as} = 2.823 \text{ Hz}$. Table 1 lists the vertical and torsional structural damping ratios ξ_{hs} and ξ_{as} of different cases.

3. Characteristics of vibration amplitude and frequency

Fig. 3 displays the time history examples of displacements from rest to a steady-state amplitude at different wind speed regions at angle of attack $\alpha = 3^\circ$ with damping ratios of Case 1 in Table 1. These responses correspond to vertical VIV, torsional VIV and coupled flutter, respectively. Fig. 4 displays the steady-state amplitudes of vertical and torsional displacements at different wind speeds and angles of attack. Fig. 5 shows the variations of vertical and torsional frequencies. The following three different types of vibrations are observed: (1) Vertical VIV around wind speed $U = 1.0 \text{ m/s}$, with a reduced wind speed of $U/(f_{hs}B) = 1.23$. The

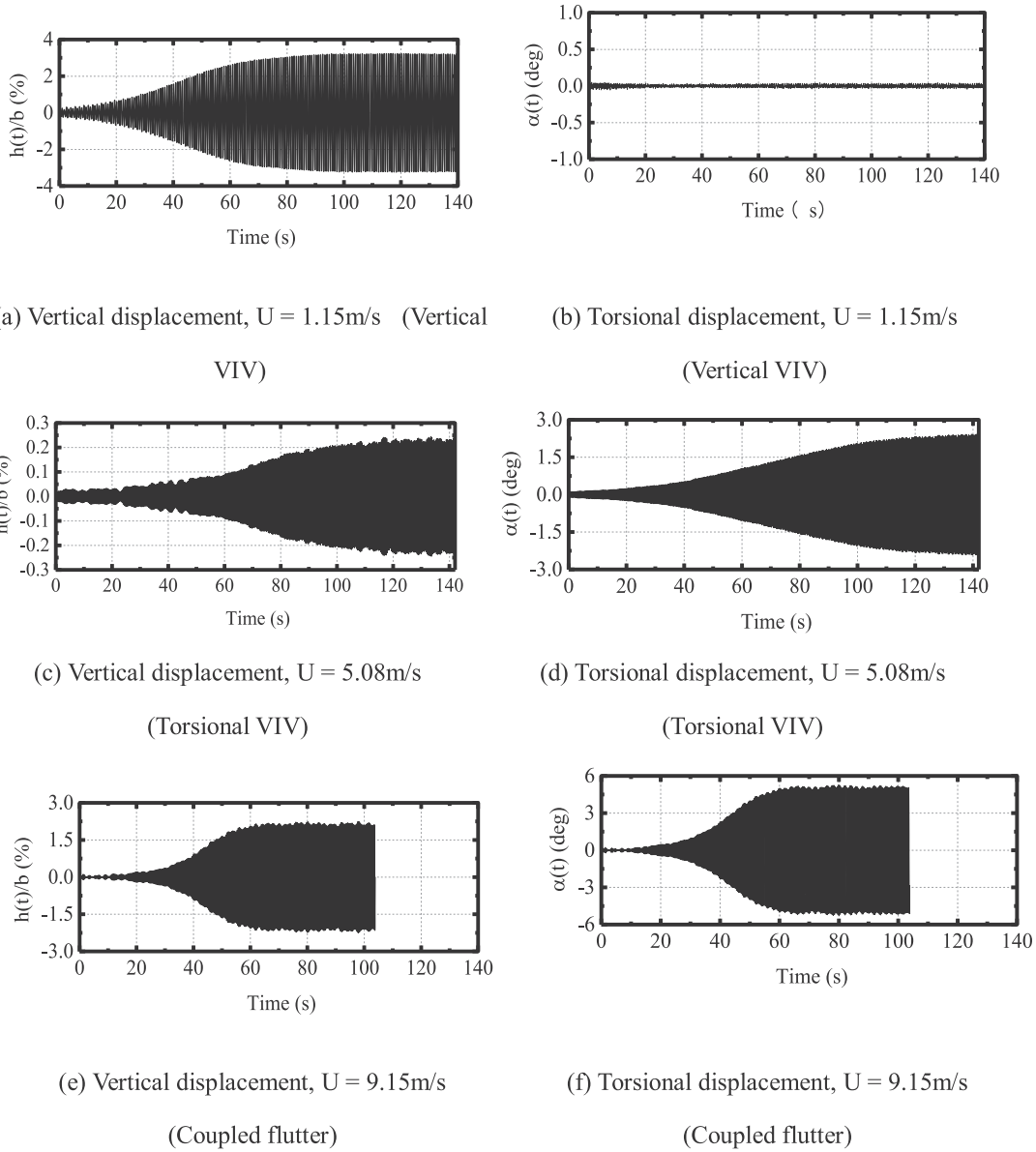


Fig. 3. Time history examples of displacements for vertical and torsional VIVs and coupled flutter ($\alpha = 3^\circ$, Damping case 1).

vibration frequency is same as the vertical mechanical frequency of the system at zero wind speed; (2) Torsional VIV around wind speed $U = 4.8$ m/s, with a reduced wind speed of $U/(f_{as}B) = 2.75$. Slightly coupled vertical motion is also observed at angles of attack of 3° , 5° and 7° . The vibration frequencies of both vertical and torsional motion is very close to the torsional mechanical frequency of the system; (3) Flutter with coupled vertical and torsional vibrations at higher wind speed. Both vertical and torsional motions have same frequency that is very close to the torsional mechanical frequency of the system. It is noted that the wind speed ranges of vertical and torsional VIVs remain almost unaffected when the angle of attack varies, while the vibration amplitude is sensitive to the angle of attack. On the other hand, the flutter onset wind speed decreases with the increase of angle of attack. As a result, the torsional VIV and flutter mixed together at large angle of attack and lower damping ratio. For a wide range of wind speed, especially at the case of larger angle of attack, the flutter leads to coupled limit cycle oscillations which indicates the existence of nonlinearity of the system. The effect of structural damping on vibration amplitude varies for different types of vibration at different angles of attack, which is associate with the nonlinear dependence of aerodynamic damping on vibration amplitude.

At the steady-state amplitude, the negative aerodynamic damping cancels out the positive structural dynamic so the system has a zero damping. It should be noted that at wind speed ranging from around 1.5 to 3 m/s, there is no self-excited vibration observed. At zero angle of attack, the vibration amplitude at the wind speed ranging from around 6 to 8 m/s is also very low and thus is not further investigated.

4. Uncoupled VIV IN vertical direction

4.1. Extraction of aerodynamic damping

The vertical VIV is treated as a pure vertical motion. Fig. 6 shows the time history of vertical displacement at $U = 1.15$ m/s, $\xi_{hs} = 0.32\%$ and $\alpha = 0^\circ$. The vibration grows with time from rest to a steady-state response.

To investigate the amplitude-dependent damping ratio, the time-varying amplitude $A_h(t)$ is obtained through Hilbert Transform (HT) of the response (i.e., Hahn, 1996; Wu and Chen, 2019). Its natural logarithm is then fitted into the following analytical function, known as Gompertz model, to facilitate the calculation of its derivative with respect to

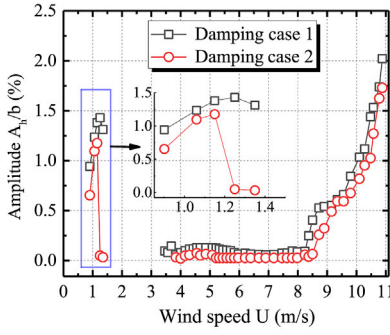
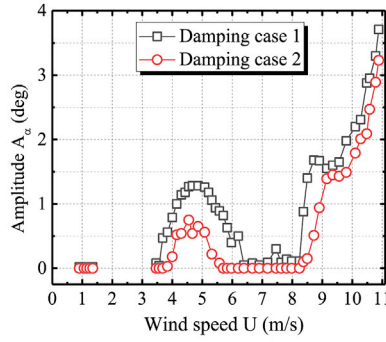
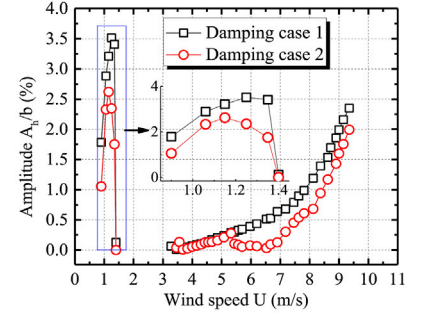
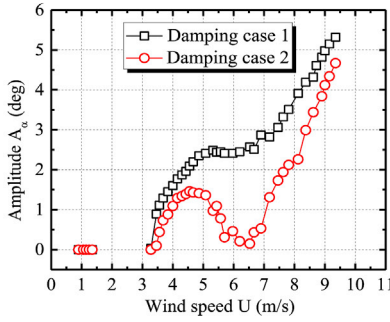
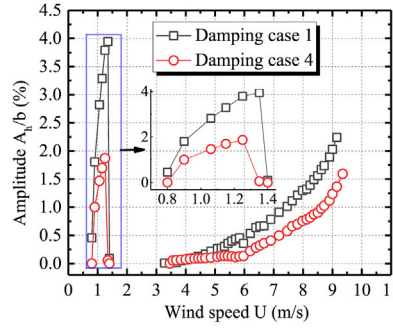
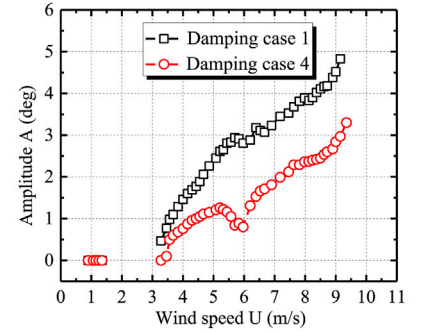
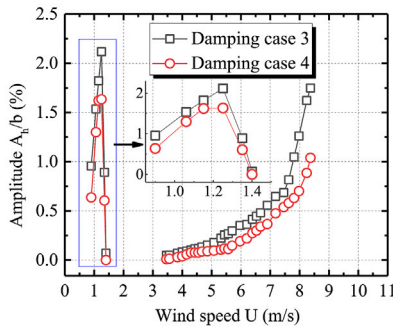
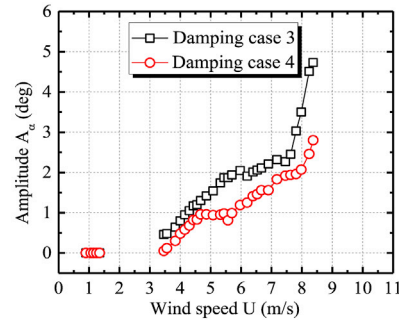
(a) Vertical displacement ($\alpha = 0$ deg)(b) Torsional displacement ($\alpha = 0$ deg)(c) Vertical displacement ($\alpha = 3$ deg)(d) Torsional displacement ($\alpha = 3$ deg)(e) Vertical displacement ($\alpha = 5$ deg)(f) Torsional displacement ($\alpha = 5$ deg)(g) Vertical displacement ($\alpha = 7$ deg)(h) Torsional displacement ($\alpha = 7$ deg)

Fig. 4. Vibration amplitudes vs. wind speed.

time. The Gompertz model was selected mainly because it has a similar trend compared to the time-varying vibration amplitude.

$$\ln\left(\frac{A_h}{A_{hm}}\right) = a_0 \exp\{-\exp[a_1(t - t_c)]\} \quad (1)$$

where A_{hm} is the steady-state amplitude; a_0 and a_1 are constants; t_c is the time instant corresponding to $\ln(A_h/A_{hm}) = 0.368a_0$, which represents the inflection point of the logarithmic amplitude envelope. The fitted amplitude of the response time history is also shown in Fig. 6, which perfectly matches the amplitude data. The vibration frequency can also be obtained from HT, which can be treated as time-invariant and $f_h = f_{hs} = 1.294$ Hz.

The damping ratio is calculated as

$$\xi_h(t) = -\frac{1}{2\pi f_h} \frac{d\ln(A_h)}{dt} \quad (2)$$

The amplitude-dependent damping ratio is then calculated as

$$\xi_h(A) = \frac{a_1}{2\pi f_h} \ln\left(\frac{A_h}{A_{hm}}\right) \ln\left[\frac{a_0}{\ln(A_h) - \ln(A_{hm})}\right] \quad (3)$$

Fig. 7 displays the total damping ratio and aerodynamic damping ratio, as functions of time and amplitude of vertical displacement, respectively. The aerodynamic damping is obtained by subtracting the structural damping from the total damping. It is evident that the system is subjected to negative damping at lower amplitudes. The energy is continuously absorbed during the vibration so that the vibration amplitude gradually increases until the total damping ratio reaches zero value and the vibration reaches a steady-state amplitude. The steady-state

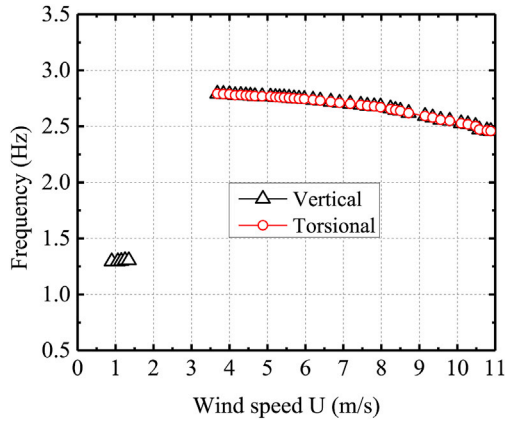
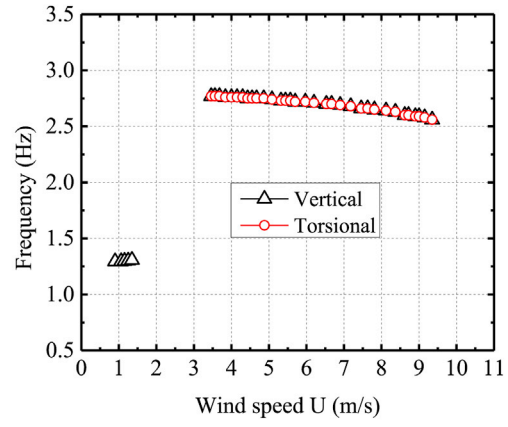
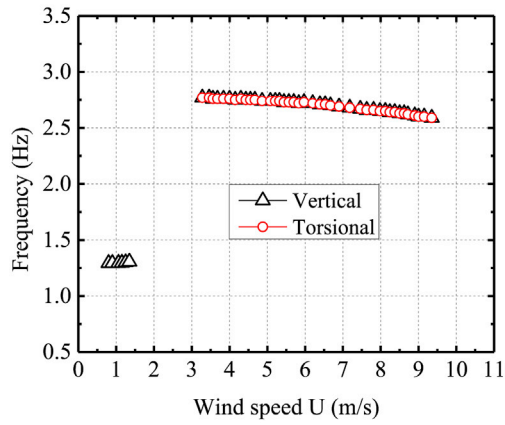
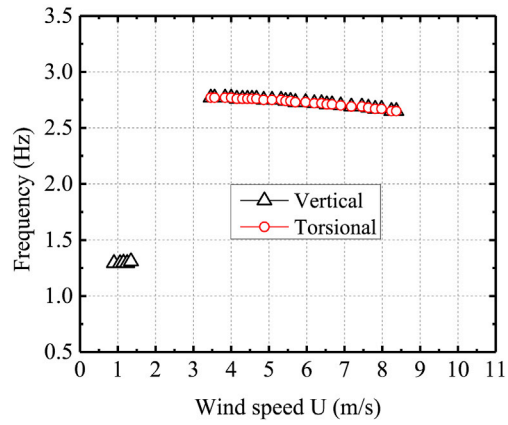
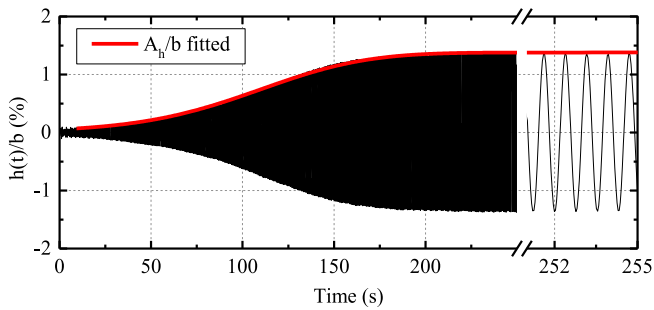
(a) $\alpha = 0$ deg, Damping case 1(b) $\alpha = 3$ deg, Damping case 1(c) $\alpha = 5$ deg, Damping case 3(d) $\alpha = 7$ deg, Damping case 3

Fig. 5. Vibration frequencies vs. wind speed.

Fig. 6. Time history of vertical displacement ($\alpha = 0$ deg, $U = 1.15$ m/s, $\xi_{hs} = 0.32\%$).

amplitude is $A_h = 4.33$ mm, i.e., $A_h/b = 1.38\%$ at $U = 1.15$ m/s.

Fig. 8 displays the amplitude-dependent aerodynamic damping associated with vertical VIV at different wind speeds. At $U = 1.06$ m/s, the amplitude of negative aerodynamic damping is maximum. With the amplitude-dependent damping, the steady-state amplitude of VIV at $\xi_{hs} = 0.32\%$ and 0.45% can be readily calculated and then compared with the measurement data as shown in Fig. 9. As expected, the prediction and experimental data at $\xi_{hs} = 0.32\%$ are identical. The predictions at $\xi_{hs} = 0.45\%$ are in good agreement with the experimental data except at wind speed $U = 1.25$ m/s, where the negative aerodynamic damping is very

low and only slowly varies with amplitude as shown in Fig. 8.

The aerodynamic damping is also calculated from the response at $\xi_{hs} = 0.45\%$. The results are shown in Fig. 10. The maximum absolute error of aerodynamic damping ratios of these two experiments are less than 0.1% . The results indicate that the aerodynamic damping is the inherent aerodynamic character of the bridge deck. The accuracy of the derived aerodynamic damping is confirmed.

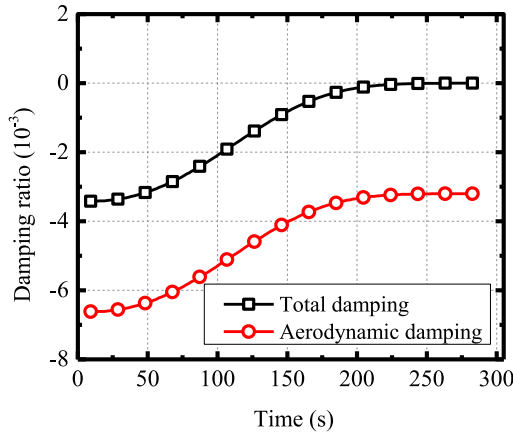
4.2. Extraction of aerodynamic derivatives

The equation of VIV in vertical SDOF of the bridge deck can be written as

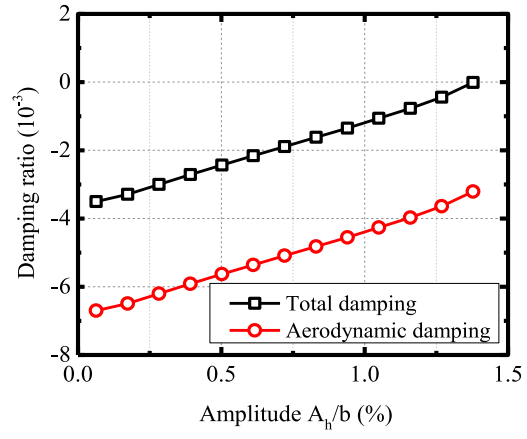
$$m(\ddot{h} + 2\xi_{hs}\omega_{hs}\dot{h} + \omega_{hs}^2 h) = \frac{1}{2}\rho U^2(2b) \left(kH_1^* \frac{\dot{h}}{U} + k^2 H_4^* \frac{h}{b} \right) \quad (4)$$

where h is the vertical displacement of bridge deck; ρ is air density; $b = B/2$ is the half width of the bridge deck; $k = \omega b/U$ is reduced frequency; ω is circular frequency of vibration; H_1^* and H_4^* are flutter (aerodynamic) derivatives related to the aerodynamic damping and stiffness respectively, which are functions of reduced frequency and amplitude.

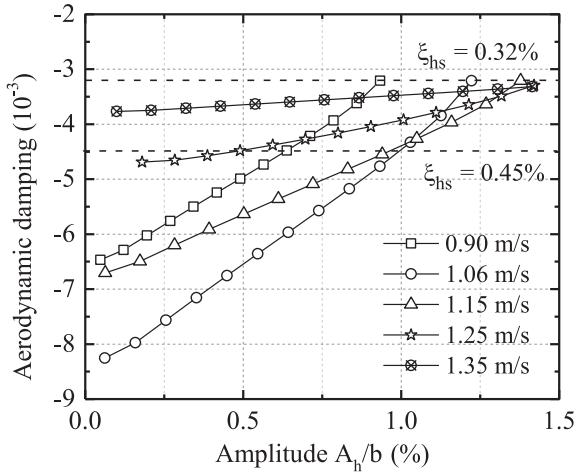
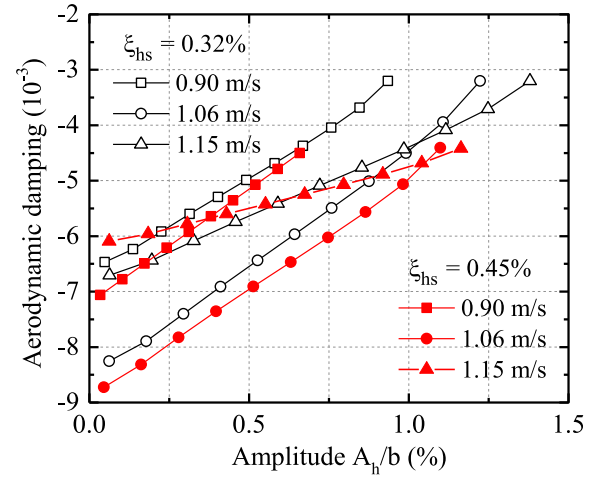
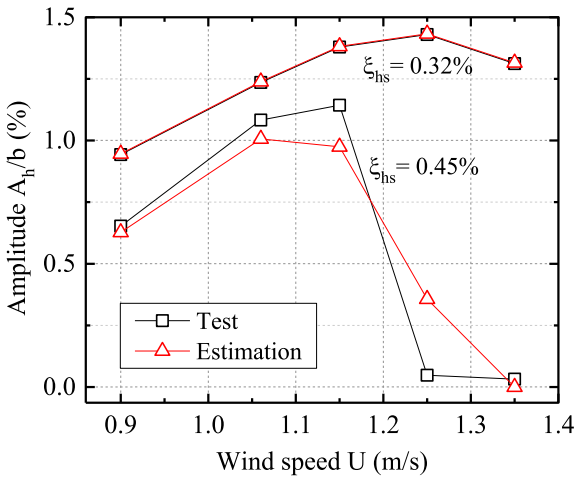
For vertical VIV, the influence of aerodynamic stiffness H_4^* on structural frequency is negligible small, i.e., $\omega_h \approx \omega_{hs}$. The aerodynamic damping ratio is related to H_1^* as:



(a) Damping ratio vs. time



(b) Damping ratio vs. amplitude

Fig. 7. Vertical damping ratio ($\alpha = 0$ deg, $U = 1.15$ m/s, $\xi_{hs} = 0.32\%$).Fig. 8. Vertical aerodynamic damping ratio ($\xi_{hs} = 0.32\%$, $\alpha = 0$ deg).Fig. 10. Comparison of vertical aerodynamic damping ratios ($\alpha = 0$ deg).Fig. 9. Comparison of amplitude from estimation and test ($\alpha = 0$ deg).

$$\xi_{ha} = -\frac{1}{2}\mu H_1^* \quad (5)$$

where $\mu = \rho b^2/m$ is mass ratio.

According to [Chen \(2013\)](#), the aerodynamic derivative H_1^* can be expressed as a polynomial function of dimensionless amplitude

$$H_1^* = B_1 + \frac{4}{3\pi}B_2\left(\frac{A_h}{b}\right) + \frac{1}{4}B_3\left(\frac{A_h}{b}\right)^2 \quad (6)$$

which is equivalent to the following model in terms of time-varying displacement:

$$H_1^* = B_1 + B_2\left|\frac{h}{b}\right| + B_3\left(\frac{h}{b}\right)^2 \quad (7)$$

In [Ehsan and Scanlan \(1990\)](#), the self-excited lift force associated with VIV is given as:

$$L_{se}(t) = \frac{1}{2}\rho U^2(2B)\left[Y_1(k)\left(1 - \varepsilon\frac{h^2}{B^2}\right)\frac{\dot{h}}{U} + Y_2(k)\frac{h}{B}\right] \quad (8)$$

where ε is a constant; and $Y_1(k)$ is aerodynamic derivative. Obviously,

$$H_1^* = \frac{2Y_1(k)}{k}\left(1 - \varepsilon\frac{h^2}{B^2}\right) \quad (9)$$

Comparing with Eq. (9), we have $B_1 = \frac{2Y_1k}{k} = K_{a0}k$, $B_2 = 0$, $B_3 = \frac{1}{4}\varepsilon K_{a0}$. It should be mentioned that ε can vary with wind speed.

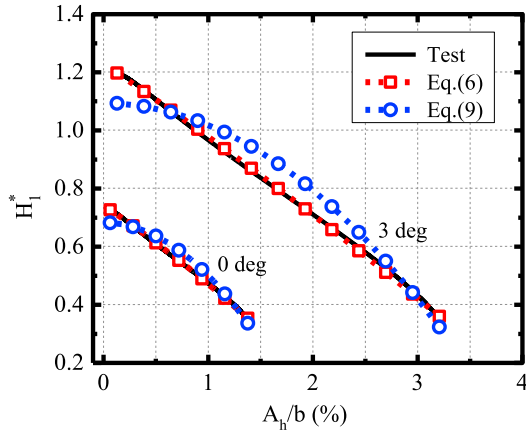
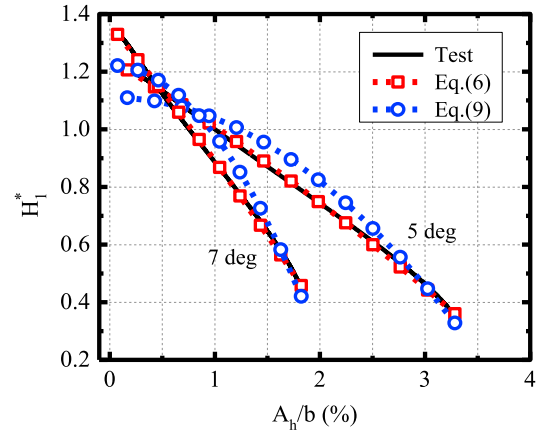
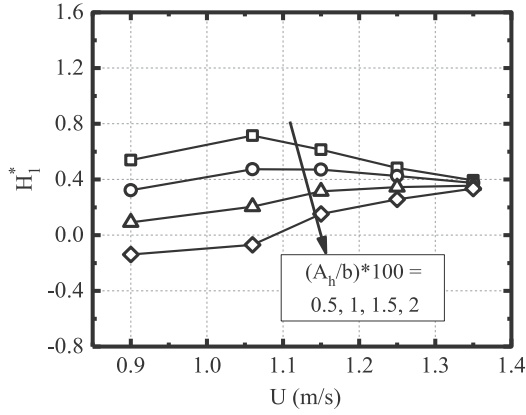
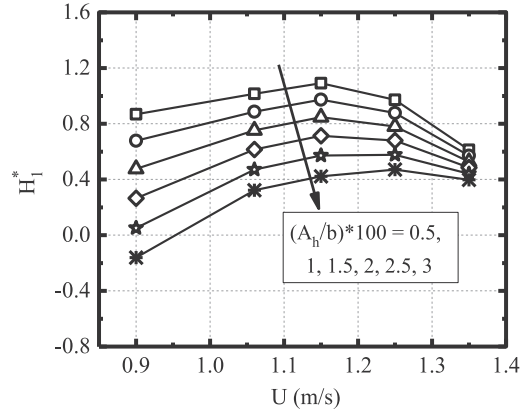
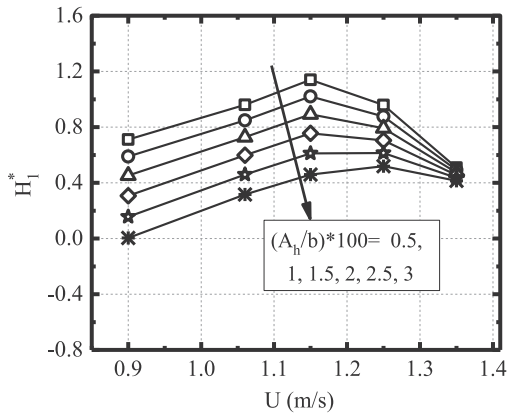
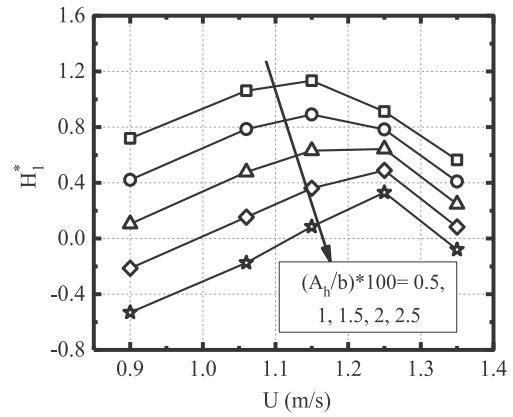
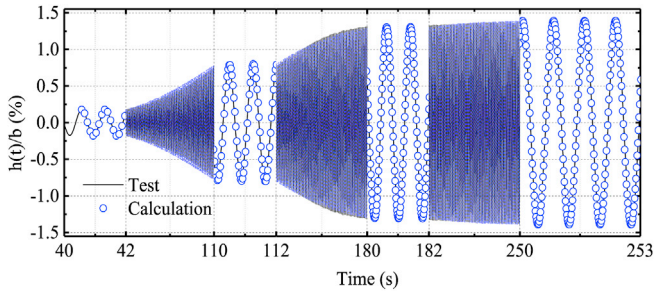
(c) $\alpha = 0$ and 3 deg(d) $\alpha = 5$ and 7 degFig. 11. H_1^* fitted from two models ($U = 1.15$ m/s).(a) $\alpha = 0$ deg(b) $\alpha = 3$ deg(c) $\alpha = 5$ deg(d) $\alpha = 7$ degFig. 12. Aerodynamic derivative H_1^* vs. wind speed and amplitude.

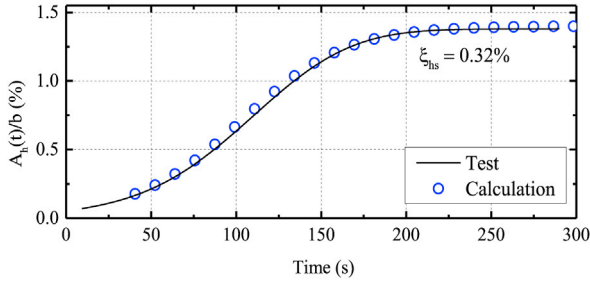
Fig. 11 shows H_1^* at $U = 1.15$ m/s fitted as a function of dimensionless vertical amplitude by using these two models, i.e., Eq. (6) introduced in Chen (2013) with linear term A_h/b , and Eq. (9) introduced in Ehsan and Scanlan (1990). As expected, with the additional linear term, the amplitude dependence of aerodynamic damping can be better captured.

In fact, higher order polynomials can also be added as needed.

Fig. 12 shows H_1^* varying with wind speed and dimensionless amplitude A_h/b at different angles of attack. H_1^* decreases gradually as the amplitude increases. The magnitude of H_1^* first increases with the wind speed and then decreases, and reaches its maximum at wind speed

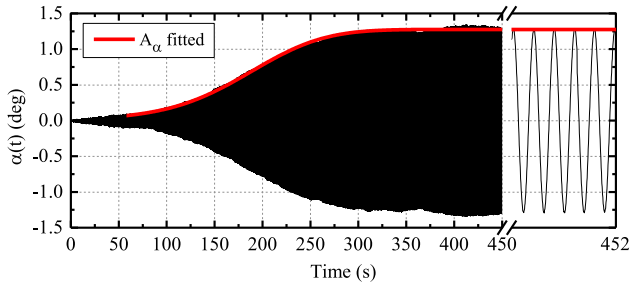


(a) Response time history

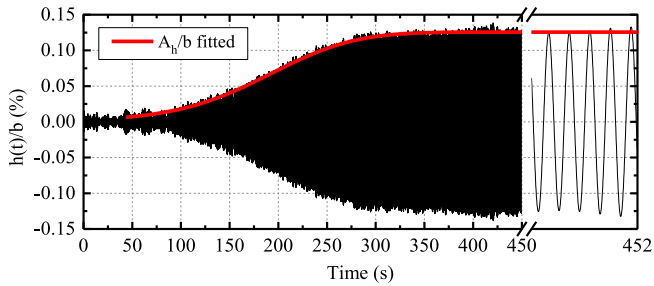


(b) Time-varying amplitude

Fig. 13. Comparison of vertical displacement and time-varying amplitude ($\alpha = 0$ deg, $U = 1.15$ m/s, $\xi_{hs} = 0.32\%$).



(a) Torsional displacement



(b) Vertical displacement

Fig. 14. Time history of bridge deck displacement ($\alpha = 0$ deg, $U = 4.67$ m/s, damping case 1).

$U = 1.15$ m/s. The sensitivity of H_1^* to amplitude reduces at higher wind speeds that are away from the vortex lock-in speed. It should be mentioned that the results at amplitudes greater than that of test are extrapolated from the curve-fitted models.

The nonlinear flutter derivative H_1^* can be used to calculate the time history of vertical displacement by solving the equation of motion via

Runge Kutta method. The time-varying amplitude can also be directly computed by solving the following equation (Feng and Chen, 2018):

$$\dot{A}_h(t) = -(\xi_{hs} + \xi_{ha}(A_h))A_h(t) \quad (10)$$

and the steady-state amplitude is determined by setting $\dot{A}_h(t) = 0$, which leads to $\xi_{hs} + \xi_{ha}(A_h) = 0$, i.e., zero total system damping.

Fig. 13 shows the results at $U = 1.15$ m/s, $\alpha = 0$ deg, and $\xi_{hs} = 0.32\%$, and compared with the experimental data. The initial conditions are $t_0 = 40.563$ s, $h_0/b = 0.178\%$ and $\dot{h}_0 = 0$. Both the response time history and the time-varying amplitude perfectly match the wind tunnel data.

5. VIV and flutter with coupled vibrations

5.1. Response characteristics

Fig. 14 displays the time histories of coupled vertical and torsional displacements at $U = 4.67$ m/s, $\alpha = 0^\circ$ and damping case 1. The HT is carried out for both vertical and torsional displacements, which gives the instantaneous frequencies, amplitudes and phases. The results are moving averaged over the time window of $1/f_{as}$, where f_{as} is torsional frequency of the system. Fig. 15 shows the frequencies, amplitude ratio $A_h(t)/bA_\alpha(t)$ and phase difference $\vartheta_h(t) - \vartheta_\alpha(t)$. The response at beginning period of time, say, when $A_\alpha(t) < 0.4^\circ$, involves contributions of two modes. Therefore, the HT analysis results cannot be used. The HT analysis after $T = 150$ s, i.e., $A_\alpha(t) > 0.4^\circ$, shows that both vertical and torsional displacements have identical time-invariant frequency. The phase difference is positive, i.e., the vertical displacement (downward positive) leads the torsional displacement (nose-up positive). The amplitude ratio and phase difference only slightly vary with time or vibration amplitude, and can be considered as amplitude independent for simplicity.

5.2. Extraction of aerodynamic damping

The amplitude-dependent damping is calculated from the time-varying vertical and torsional vibration amplitudes following previously described procedure. As shown in Fig. 16, the damping ratio derived from both displacements can be considered identical by neglecting slight numerical errors. The aerodynamic damping ratio is then obtained by subtracting the torsional mechanical damping from the total damping ratio. Similar analysis is carried out for different wind speeds and angles of attack.

The aerodynamic damping is fitted into a quadratic polynomial function of torsional amplitude:

$$\xi_a = B_{1a} + \frac{4}{3\pi}B_{2a}A_\alpha + \frac{1}{4}B_{3a}A_\alpha^2 \quad (11)$$

As shown in Fig. 17, with additional linear term, i.e., the term with B_{2a} , the amplitude dependence of damping can be better captured. In fact, higher order polynomials can also be added as needed. Fig. 18 shows the results at higher wind speeds at different angle of attack. The highest negative aerodynamic damping with obvious amplitude dependency is observed at $\alpha = 5^\circ$.

Fig. 19 displays the aerodynamic damping ratio as a function of both wind speed and vibration amplitude at different angles of attack. The aerodynamic damping at angle of attack of $\alpha = 0$ deg and wind speed range from around 6 to 8 m/s was not able to be determined as the self-excited vibration was too low to give meaningful result. Instead, a decaying vibration record can be measured and used to extract the aerodynamic damping but such a test was not carried out. It is observed that the aerodynamic damping decreases as the amplitude increases. At coupled VIV region, the aerodynamic damping reaches its maximum at the wind speed when the VIV amplitude is largest. At flutter region, on the other hand, the aerodynamic damping increases with increasing wind speed. The magnitude of negative aerodynamic damping is larger than

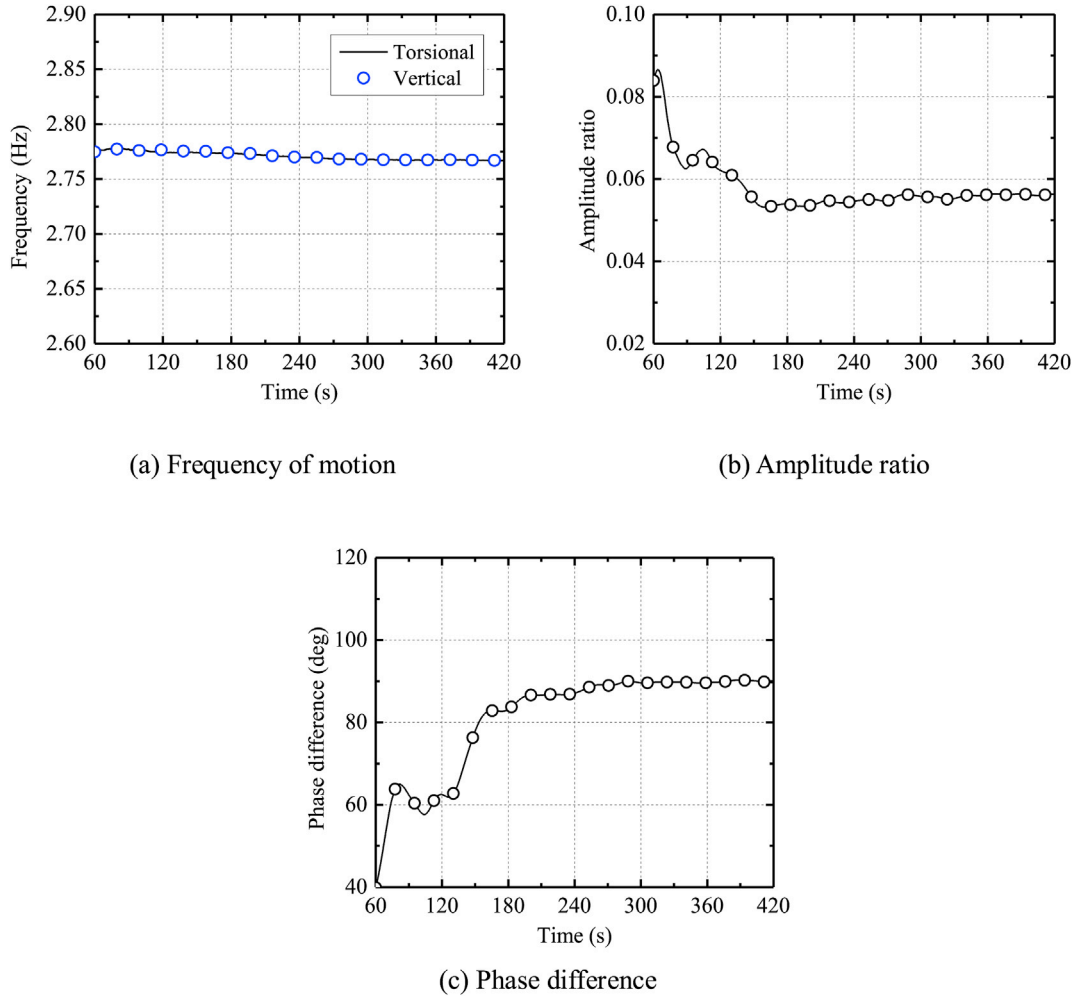


Fig. 15. Frequency, amplitude ratio and phase difference ($\alpha = 0^\circ$, $U = 4.67$ m/s, damping case 1).

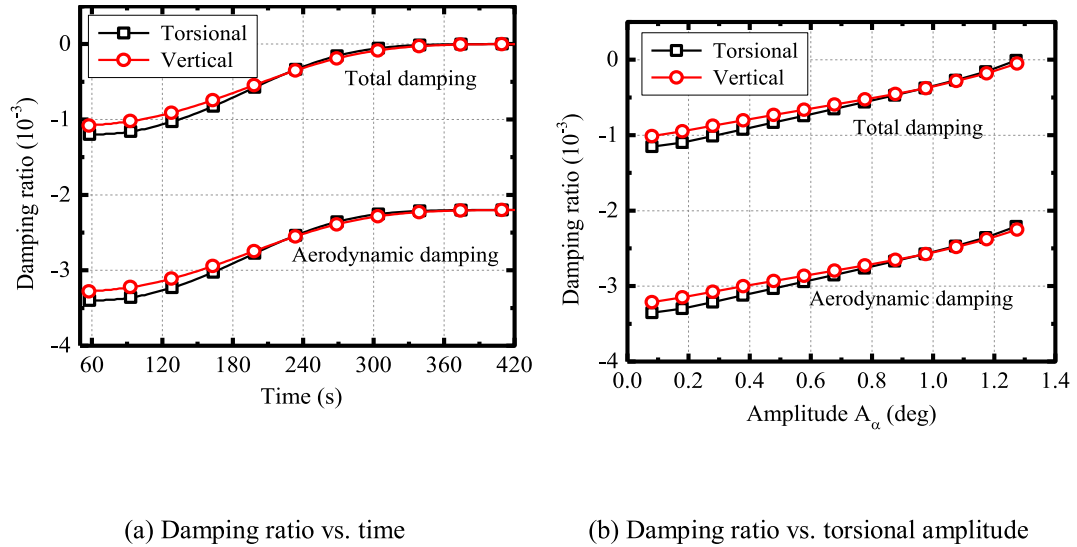


Fig. 16. Calculated damping ratios ($\alpha = 0^\circ$, $U = 4.67$ m/s, damping case 1).

that at lower VIV wind speed region. The largest aerodynamic damping is observed at angle of attack $\alpha = 5^\circ$.

Fig. 20 displays the amplitude ratio and phase difference at $\alpha = 0$ and 5° . At $\alpha = 0^\circ$, the amplitude ratio is insensitive to wind speed in the VIV

region but increases rapidly with wind speed in flutter region. On the other hand, the phase difference decreases rapidly with wind speed in the VIV region, but remains almost a constant at flutter region. At $\alpha = 5^\circ$, the amplitude ratio increases with wind speed in both VIV and flutter

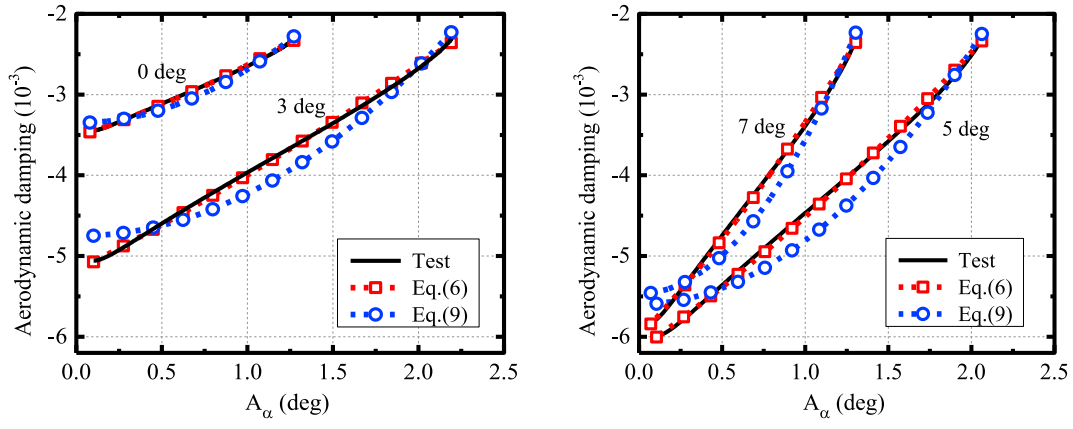
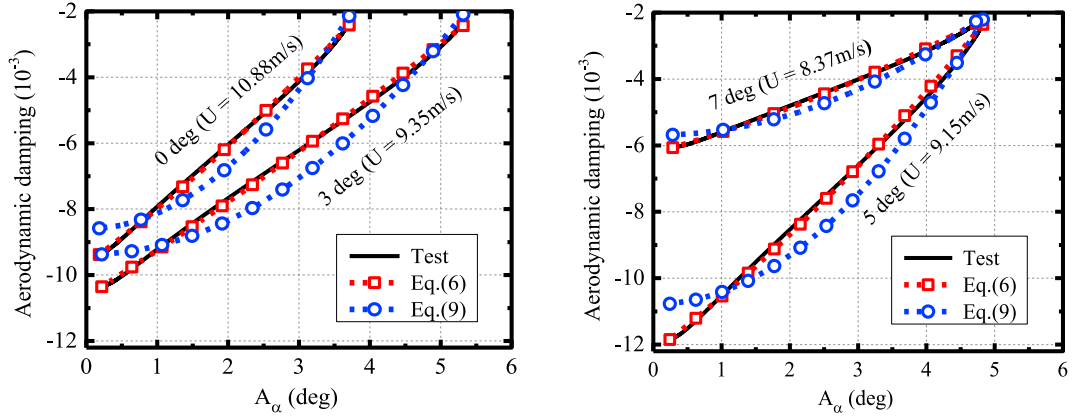
(a) $\alpha = 0$ and 3 deg(b) $\alpha = 5$ and 7 degFig. 17. Aerodynamic damping fitted as a polynomial function of amplitude ($U = 4.67$ m/s).(c) $\alpha = 0$ and 3 deg(d) $\alpha = 5$ and 7 deg

Fig. 18. Aerodynamic damping fitted as a polynomial function of amplitude (Flutter region).

regions. The phase difference slightly decreases with wind speed in the VIV region but increases at the flutter region. The phase difference at the VIV region is lower than the case with $\alpha = 0$ deg. The amplitude ratio can be considered as independent of torsional amplitude for simplicity. The dependence of phase difference on torsional amplitude is not noticeable at $\alpha=0$ deg, but can be observed at $\alpha=5$ deg particularly in the torsional VIV region, i.e., wind speed from 3.5 m/s to 6 m/s.

5.3. Extraction of flutter derivatives

The equations of motion of the 2DOFs system are given as (e.g., [Chen et al., 2004, 2006 and 2007](#)):

$$m(\ddot{h} + 2\xi_{s1}\omega_{s1}\dot{h} + \omega_{s1}^2 h) = \frac{1}{2}\rho U^2 (2b) \left(kH_1^* \frac{\dot{h}}{U} + kH_2^* \frac{b\dot{\alpha}}{U} + k^2 H_3^* \alpha + k^2 H_4^* \frac{h}{b} \right) \quad (12)$$

$$I(\ddot{\alpha} + 2\xi_{s2}\omega_{s2}\dot{\alpha} + \omega_{s2}^2 \alpha) = \frac{1}{2}\rho U^2 (2b^2) \left(kA_1^* \frac{\dot{h}}{U} + kA_2^* \frac{b\dot{\alpha}}{U} + k^2 A_3^* \alpha + k^2 A_4^* \frac{h}{b} \right) \quad (13)$$

where ω_{s1} , ω_{s2} and ξ_{s1} , ξ_{s2} are vertical and torsional structural circular frequencies and damping ratios at zero wind speed; H_i^* and A_i^* ($i = 1-4$) are flutter derivatives, which are functions of reduced frequency.

The modal frequency, damping ratio, amplitude ratio and phase

difference of each modal branch at a given wind speed can be determined via the solution of complex eigenvalue problem based on the known flutter derivatives. [Chen and Kareem \(2006\)](#) and [Chen \(2007\)](#) developed closed-form solution.

The contribution of vertical modal branch is rapidly damped out due to large value of modal damping. Therefore, only the contribution of torsional modal branch is retained in the measured vibration displacements at a given wind speed. The modal properties in terms of the modal frequency ω_2 , damping ratio ξ_2 , amplitude ratio Ψ and phase difference ψ derived from vibration data are not sufficient to determine eight flutter derivatives uniquely. Therefore, it is necessary to introduce approximate conditions. Firstly, as the influence of H_4^* and A_4^* are negligibly small thus it can be assumed that $H_4^* = A_4^* = 0$. Secondly, the following approximate relationship between flutter derivatives introduced in [Matsumoto et al. \(1995\)](#) is used:

$$H_1^* = kH_3^*; \quad A_1^* = kA_3^* \quad (14)$$

By setting $h(t) = A_0 b \Psi e^{i\psi} e^{i\lambda t}$, $\alpha(t) = A_0 e^{i\lambda t}$, where $\lambda = -\xi_2 \omega_2 + i\omega_2 \sqrt{1 - \xi_2^2}$; A_0 is the complex-valued constant; and $i = \sqrt{-1}$, Eqs. (12) and (13) are rewritten as

$$\mu \omega_2^2 [iH_2^* + (1 + \bar{\lambda} \Psi e^{i\psi} H_3^*)] = (\lambda^2 + 2\lambda \xi_{s1} \omega_{s1} + \omega_{s1}^2) \Psi e^{i\psi} \quad (15)$$

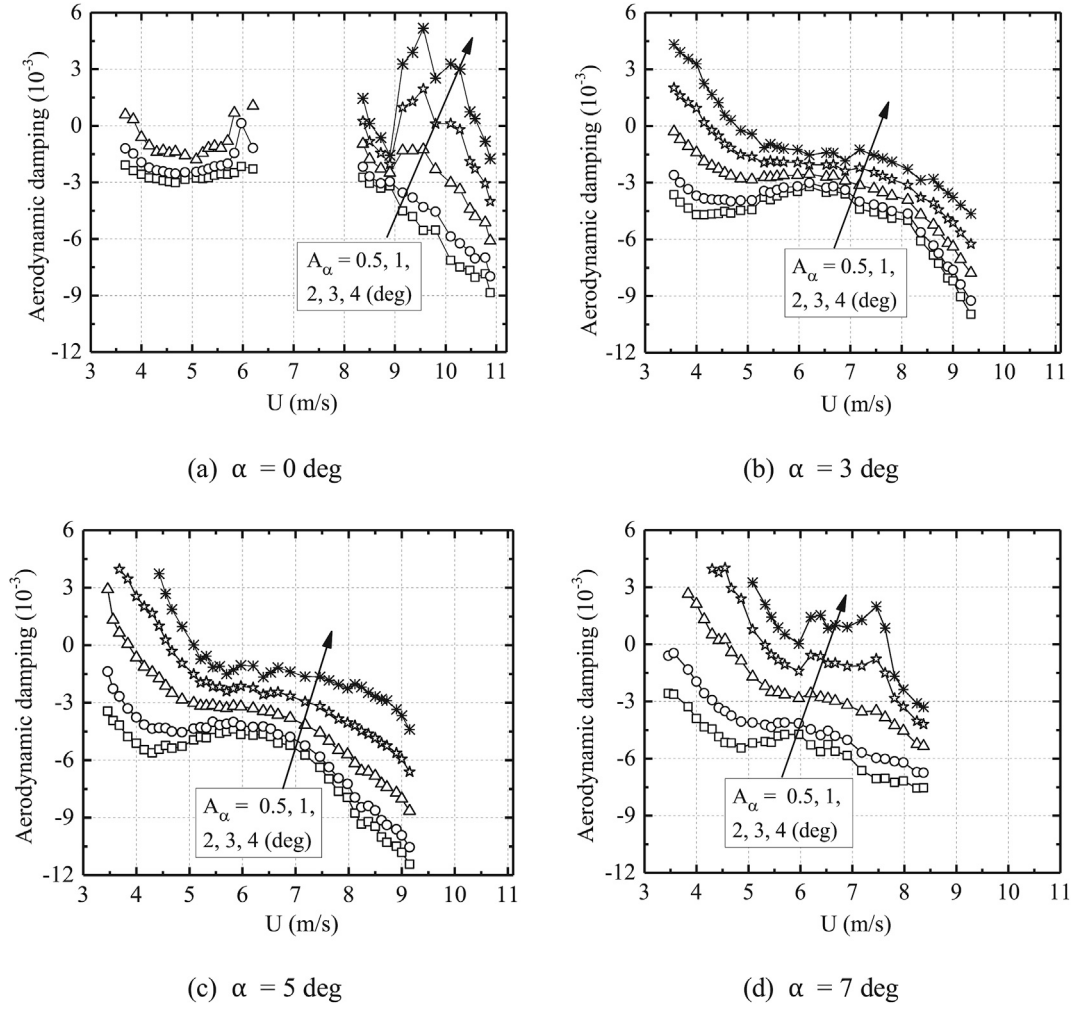


Fig. 19. Amplitude-dependent aerodynamic damping ratio vs. wind speed.

$$\nu \omega_2^2 [iA_2^* + (1 + \bar{\lambda} \Psi e^{i\psi} A_3^*)] = (\lambda^2 + 2\lambda \xi_{s2} \omega_{s2} + \omega_{s2}^2) \quad (16)$$

where $\nu = \rho b^4 / I$; and $\bar{\lambda} = \lambda b / U$.

By quantifying the real and imaginary parts on both sides of equations, applying $\bar{\lambda} \approx ik$ and neglecting higher damping terms, i.e., $\lambda^2 \approx -\omega_2^2 - 2i\xi_{s2}\omega_2^2$; $2\lambda\xi_{s1}\omega_{s1} \approx 2i\xi_{s1}\omega_{s1}\omega_2$; $2\lambda\xi_{s2}\omega_{s2} \approx 2i\xi_{s2}\omega_{s2}\omega_2$ (Chen and Kareem, 2006, 2007), we have:

$$H_3^* = \frac{2[\xi_2 - \xi_{s1}(\omega_{s1}/\omega_2)]\Psi \sin\psi - [1 - (\omega_{s1}/\omega_2)^2]\Psi \cos\psi}{\mu(1 - k\Psi \sin\psi)} \quad (17)$$

$$H_2^* = -k\Psi \cos\psi H_3^* + \frac{1}{\mu} \{ 2[\xi_{s1}(\omega_{s1}/\omega_2) - \xi_2]\Psi \cos\psi - [1 - (\omega_{s1}/\omega_2)^2]\Psi \sin\psi \} \quad (18)$$

$$A_3^* = \frac{(\omega_{s2}/\omega_2)^2 - 1}{\nu(1 - k\Psi \sin\psi)} \quad (19)$$

$$A_2^* = -k\Psi \cos\psi A_3^* + 2[\xi_{s2}(\omega_{s2}/\omega_2) - \xi_2] / \nu \quad (20)$$

It should be mentioned that the above expressions actually can also be directly derived from the closed-form solutions of complex modal properties (Chen and Kareem, 2006, 2007).

Fig. 21 portrays the flutter derivatives H_2^* , H_3^* , A_1^* and A_3^* at $\alpha = 0$ degree. H_2^* and H_3^* are slightly affected by vibration amplitude. A_3^* is directly related to the change of modal frequency and thus is independent

of vibration amplitude. A_2^* is sensitive to vibration amplitude. In the torsional VIV region, A_2^* is positive and decreases with increasing amplitude. In the flutter region, however, A_2^* becomes negative and increases rapidly as wind speed and amplitude increase.

Fig. 22 shows the flutter derivative A_2^* and H_2^* at $\alpha = 3, 5$ and 7° . As compared to the value at $\alpha = 0^\circ$, A_2^* turns to positive in a lower vibration amplitude at flutter region, indicating a negative damping effect caused by torsional vibration. Accordingly, the flutter is in fact a torsional flutter while coupled vertical motion is also generated by the coupled force self-excited components. At $\alpha = 0^\circ$, the flutter is classical coupled flutter where the negative aerodynamic damping generated by coupling of vertical and torsional motion drives the system to flutter. On the other hand, H_2^* remains negative but the magnitude is lower than that at $\alpha = 0^\circ$. The dependency of its magnitude on vibration amplitude increases with an increase in angle of attack.

5.4. Validation of flutter derivatives

The system modal damping ratio ξ_2 at a given wind speed is calculated as a function of amplitude of torsional displacement with the identified flutter derivatives using the closed-form expression (Chen and Kareem, 2006; Chen, 2007). Fig. 23 shows the contributions of different force components to the damping ratio at $U/(f_\alpha B) = 2.46$ and 3.1 and $\alpha = 0^\circ$. The uncoupled torsional motion produces negative damping. The coupled vertical motion produces very low positive damping at $U/(f_\alpha B) = 2.46$ but negative damping at higher wind speed, i.e., $U/(f_\alpha B) \geq 2.7$.

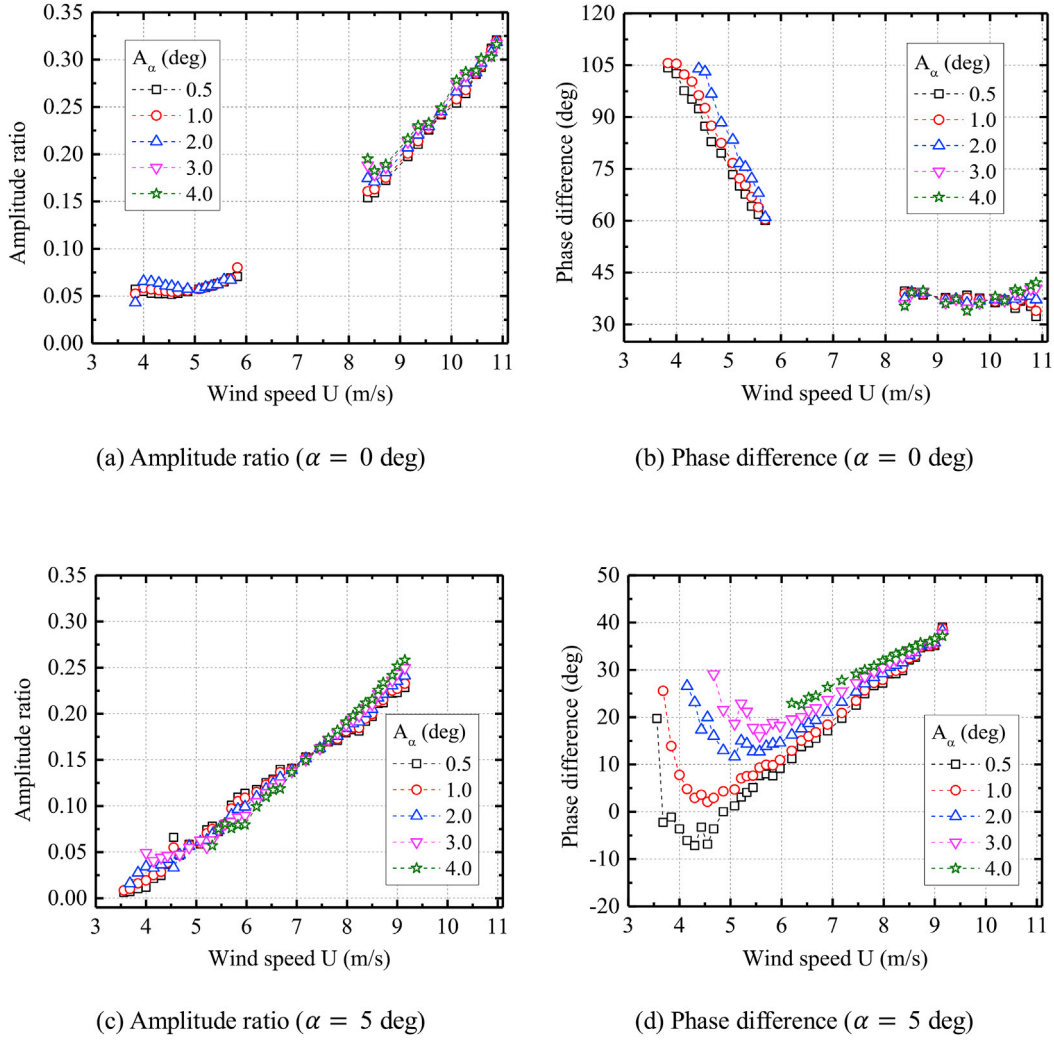


Fig. 20. Amplitude ratio and phase difference vs. wind speed and amplitude.

The closed-form estimation is not only accurate as traditional complex-eigenvalue analysis approach, but also explicitly sheds insight on the contributions of different self-excited force components on system damping.

Fig. 24 shows the contributions of different force components to the total damping at $U/fB = 5.6$ and $\alpha = 0$ and 5° . As compared with the coupled VIV, the uncoupled torsional motion at flutter region provides positive damping at $\alpha = 0^\circ$. At $\alpha = 5^\circ$, the uncoupled torsional motion provides negative damping at lower amplitude and the flutter becomes essentially a torsional flutter. The coupled vertical motion further generates negative damping.

The steady-state amplitude at a given wind speed can be calculated by setting the system damping ratio $\xi_2 = 0$. Figs. 25 and 26 present the calculated amplitudes of torsional and vertical displacements and compared with the experimental measurements at damping case 1 with $\alpha = 0$ and 5° . The very agreeable results demonstrate the accuracy of the derived flutter derivatives. The amplitude of SDOF torsional VIV is also computed, which is smaller than that of 2DOFs because the corresponding negative aerodynamic damping of SDOF system is lower. The exception is the case when $U/fB < 2.7$ and $\alpha = 0$ deg, where the coupled vertical component generates positive damping. The accuracy of the flutter derivatives is also validated by comparing the measured and predicted amplitudes at damping case 2 ($\alpha = 0^\circ$) or case 4 ($\alpha = 5^\circ$) using the derivatives obtained from damping case 1, which are also shown in Figs. 26 and 27.

The time histories of vertical and torsional displacements and their time-varying amplitudes are also computed from the identified flutter derivatives and then compared to the measurement data in order to validate the accuracy of flutter derivatives. The vertical and torsional displacements are expressed in terms of modal response as (Chen, 2002):

$$\begin{bmatrix} h(t) \\ \alpha(t) \end{bmatrix} = |A_0| e^{-\xi_2 \omega_2 t} \begin{bmatrix} b \Psi \cos \psi \cos(\omega_{d2} t + \psi + \varphi_0) \\ \cos(\omega_{d2} t + \varphi_0) \end{bmatrix} \quad (21)$$

where $|A_0| = \sqrt{\alpha_0^2 + (\dot{\alpha}_0/\omega_{d2})^2}$; $\varphi_0 = \arctan\left(\frac{\dot{\alpha}_0}{\omega_{d2} \alpha_0}\right)$; and $\omega_{d2} = \omega_2 \sqrt{1 - \xi_2^2}$.

Therefore, the relationship between $h(t)$ and $\alpha(t)$ is

$$h(t) = \Psi \cos \psi b \alpha(t) + \Psi \sin \psi b \dot{\alpha}(t) / \omega_{d2} \quad (22)$$

The time-varying amplitude of torsional displacement can also be computed from the following equation (Feng and Chen, 2018):

$$\dot{A}_\alpha(t) = -\xi_2 \omega_2 A_\alpha(t) \quad (23)$$

where the amplitude dependent damping can be accounted:

$$\xi_2 = \xi_{as} + \xi_{aa}; \quad \xi_{aa} = B_{1\alpha} + \frac{4}{3\pi} B_{2\alpha} A_\alpha + \frac{1}{4} B_{3\alpha} A_\alpha^2 \quad (24)$$

The time history of the torsional displacement can also be computed

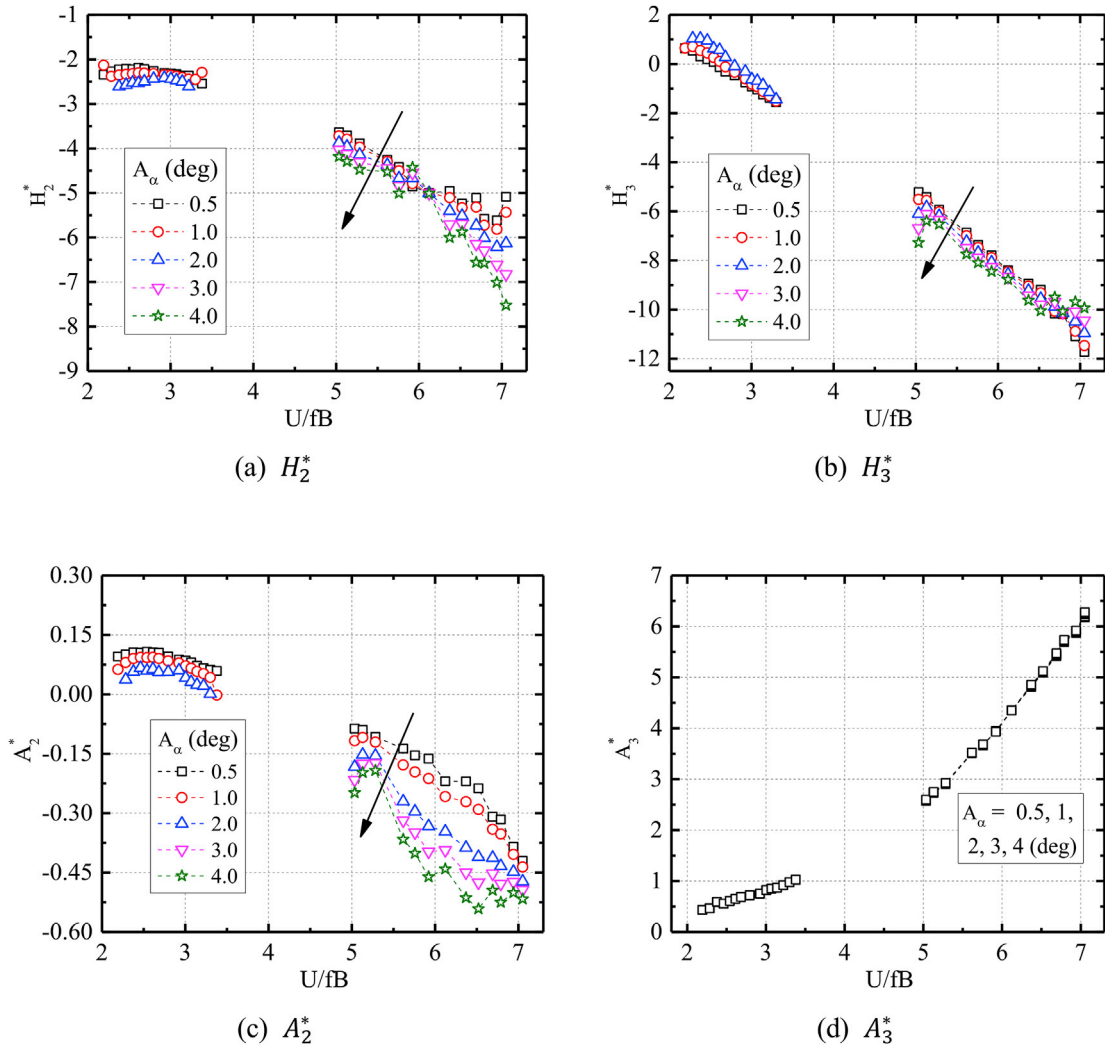


Fig. 21. Flutter derivatives at coupled VIV and flutter wind speed regions ($\alpha = 0$ deg).

from the following equation:

$$\ddot{\alpha}(t) + 2\xi_2\omega_2\dot{\alpha}(t) + \omega_2^2\alpha(t) = 0 \quad (25)$$

where the aerodynamic damping ξ_{aa} is represented by

$$\xi_{aa} = B_{1a} + B_{2a}|\alpha| + B_{3a}\alpha^2 \quad (26)$$

Fig. 27 shows the comparison of time histories of torsional and vertical displacement and time-varying amplitudes at $U = 4.67$ m/s, $\alpha = 0^\circ$ and damping case 1 with the experimental data. The initial condition is $t_0 = 164.35$ s and $\alpha_0 = 0.523$ deg and $\dot{\alpha}_0 = 0$. Both the response time history and the time-varying amplitude perfectly match the wind tunnel data. The obtained flutter derivatives are re-validated.

6. Consideration of 3D effects

6.1. Prediction of vertical VIV amplitude

In the following the vertical VIV of a full-scale long span bridge is calculated using the aforementioned amplitude-dependent self-excited force model. The 3D effects in terms of spatial variation of vibration amplitude and partial correlation of the self-excited forces at different spanwise locations will be considered in the analysis.

The displacement of VIV of a full scale long span bridge associated with j -th vertical mode can be expressed as

$$h(x, t) = h_j(x)q_j(t) \quad (27)$$

where $h_j(x)$ is mode shape; $q_j(t)$ is the generalized modal displacement.

The equation of modal displacement is (e.g., [Chen and Kareem, 2000b](#)):

$$\ddot{q}_j(t) + 2\xi_{hsj}\omega_{hsj}\dot{q}_j(t) + \omega_{hsj}^2q_j(t) = \frac{\mu}{G_{h_jh_j}} \left(\omega_{hsj}G_{h_jh_j}^{H_1^*}\dot{q}_j(t) + \omega_{hsj}^2G_{h_jh_j}^{H_4^*}q_j(t) \right) \quad (28)$$

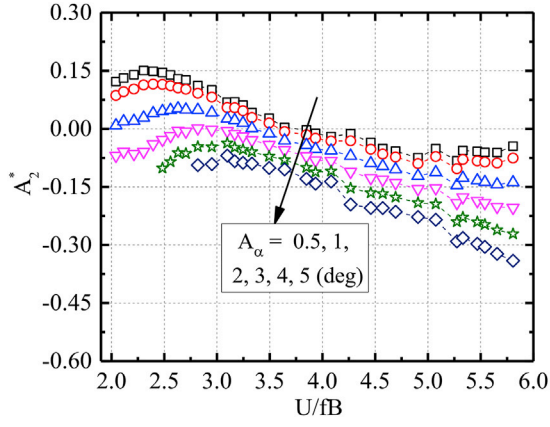
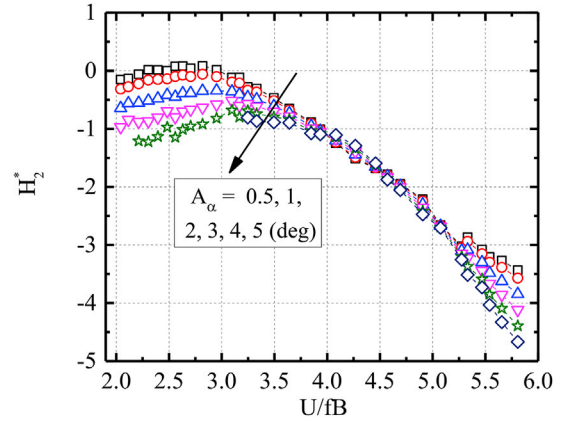
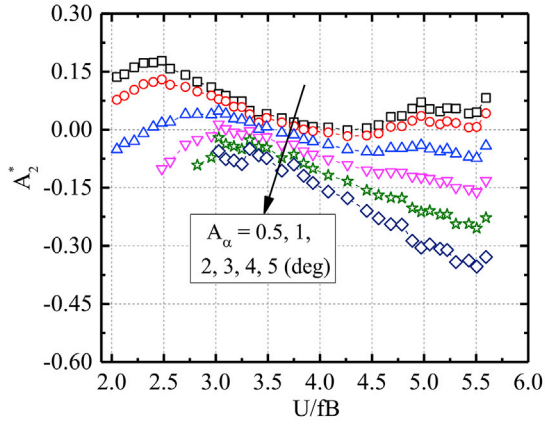
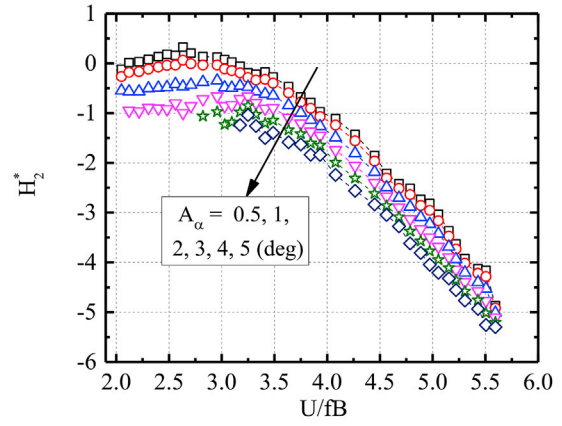
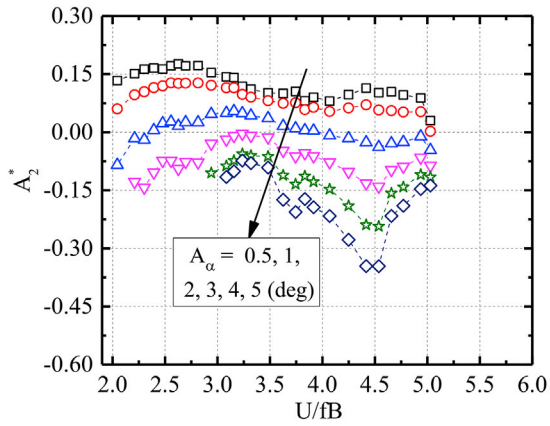
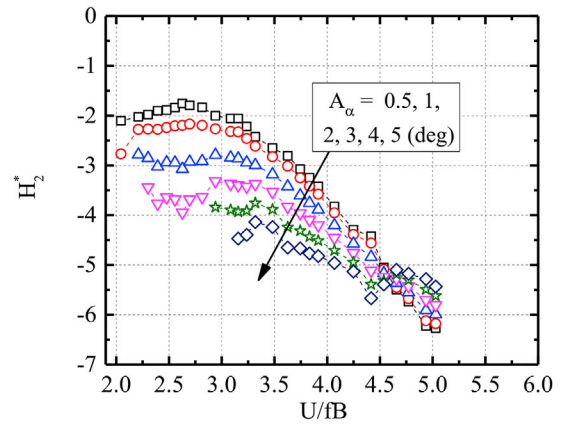
where $G_{h_jh_j} = \int_0^L h_j^2(x)dx$, $G_{h_jh_j}^{H_1^*} = \int_0^L h_j^2(x)H_1^*(x)dx$, $G_{h_jh_j}^{H_4^*} = \int_0^L h_j^2(x)H_4^*(x)dx$; and L is span length. In this 3D analysis, the amplitude-dependence of flutter derivative H_1^* and H_4^* at different spanwise location can be accounted in $G_{h_jh_j}^{H_1^*}$ and $G_{h_jh_j}^{H_4^*}$.

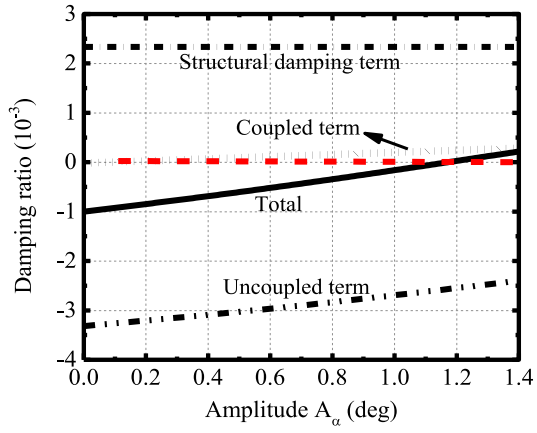
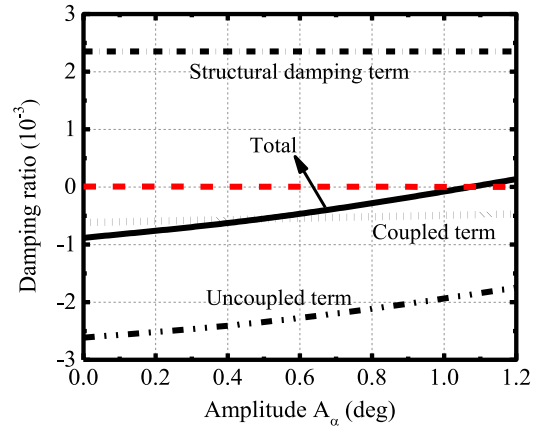
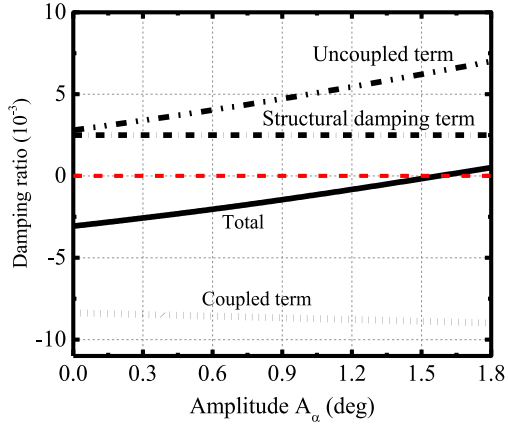
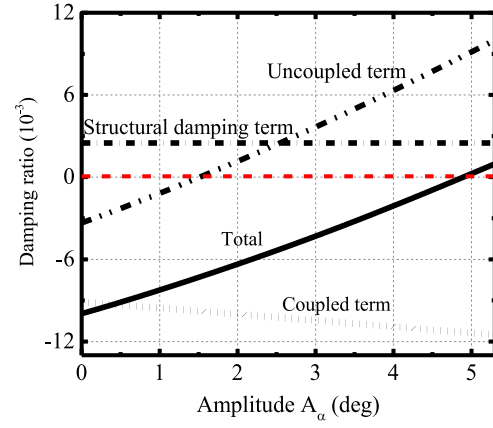
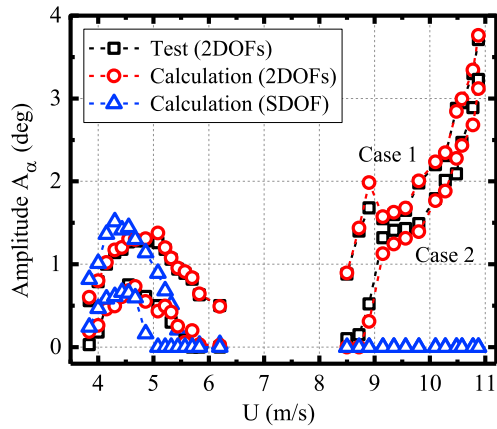
The vortex-induced forces at different spanwise locations are generally partially correlated. The correlation function of flutter derivative H_1^* can be assumed as

$$\text{cor}_{H_1^*}(x_A, x_B) = \exp\left(-c_{H_1^*}|x_A - x_B|/L\right) \quad (29)$$

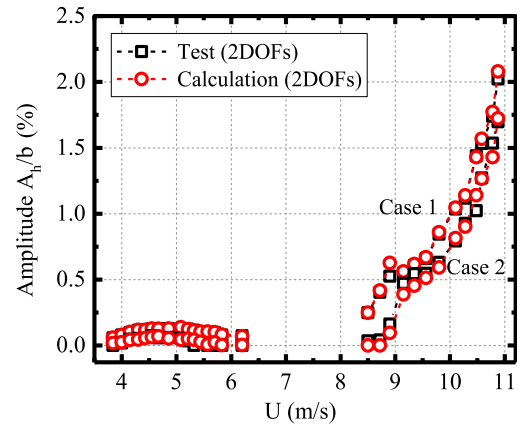
where $c_{H_1^*}$ is decay factor; x_A and x_B are two spatial positions along the bridge span.

With consideration of the correlation, $G_{h_jh_j}^{H_1^*}$ in Eq. (28) should be

(a) A_2^* , $\alpha = 3$ deg(b) H_2^* , $\alpha = 3$ deg(c) A_2^* , $\alpha = 5$ deg(d) H_2^* , $\alpha = 5$ deg(e) A_2^* , $\alpha = 7$ deg(f) H_2^* , $\alpha = 7$ degFig. 22. Flutter derivative A_2^* and H_2^* at different angles of attack.

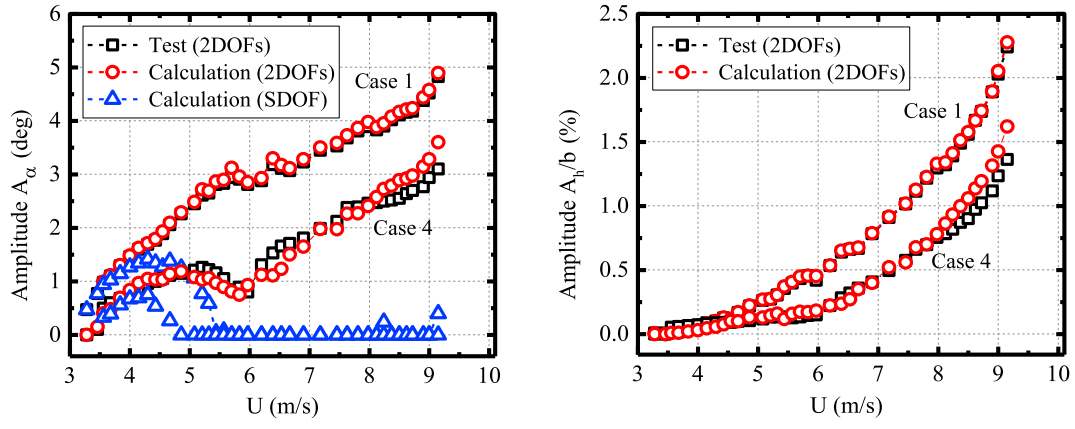
(a) $U/fB = 2.46$ (b) $U/fB = 3.2$ Fig. 23. Contributions of different force components to damping ratio (VIV region, $\alpha = 0$ deg).(a) $\alpha = 0$ deg(b) $\alpha = 5$ degFig. 24. Contributions of different force components to damping ratio (Flutter region, $U/fB = 5.6$).

(a) Torsional amplitude



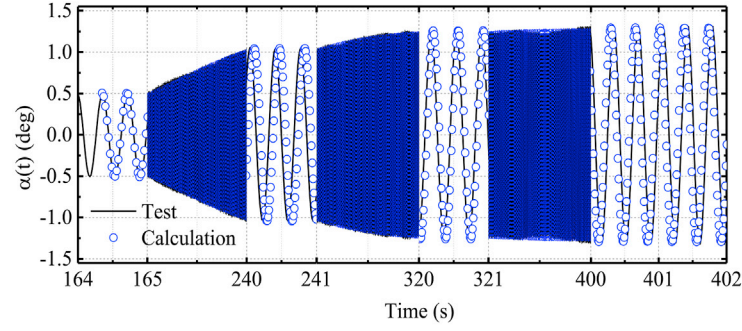
(b) Vertical amplitude

Fig. 25. Experimental and calculated steady-state amplitude vs. wind speed ($\alpha = 0$ deg).

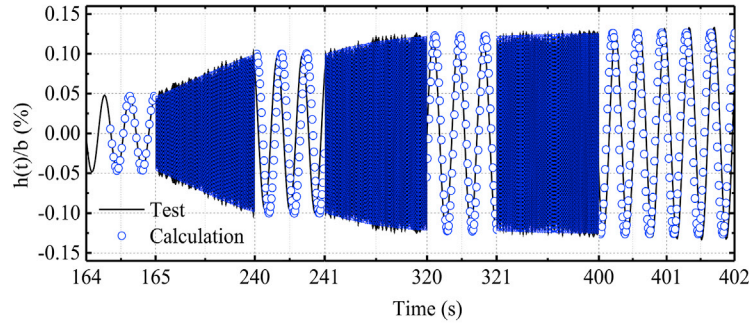


(a) Torsional amplitude

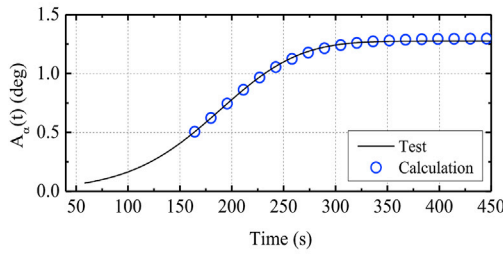
(b) Vertical amplitude

Fig. 26. Experimental and calculated steady-state amplitude vs. wind speed ($\alpha = 5$ deg).

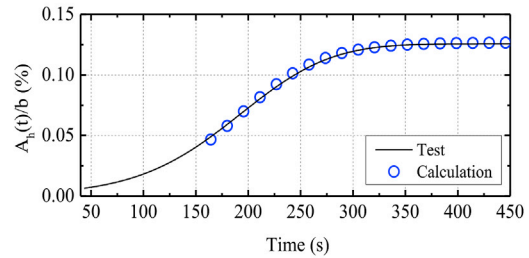
(a) Torsional displacement



(b) Vertical displacement



(c) Torsional time-varying amplitude



(d) Vertical time-varying amplitude

Fig. 27. Comparison of response time histories and time-varying amplitudes. ($\alpha = 0$ deg, $U = 4.67$ m/s, damping case 1).

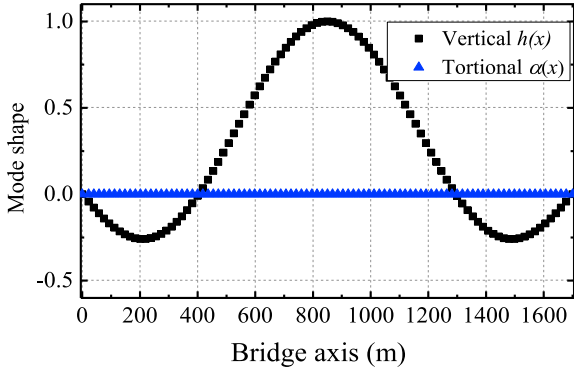
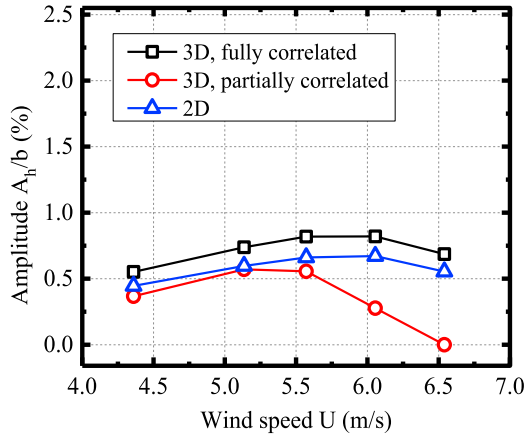


Fig. 28. Fundamental vertical mode shape of the bridge.

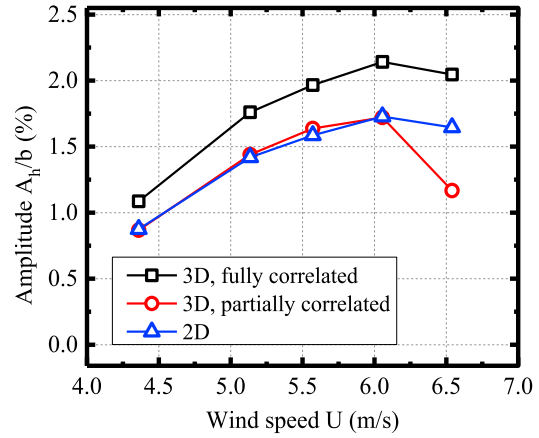
replaced by $\lambda_{H_1^*} G_{h_j h_j}^{H_1^*}$ (Mashad and Jones, 2014), where

$$\lambda_{H_1^*} = G_{c_{h_j h_j}}^{H_1^*} / G_{h_j h_j}^{H_1^*} \quad (30)$$

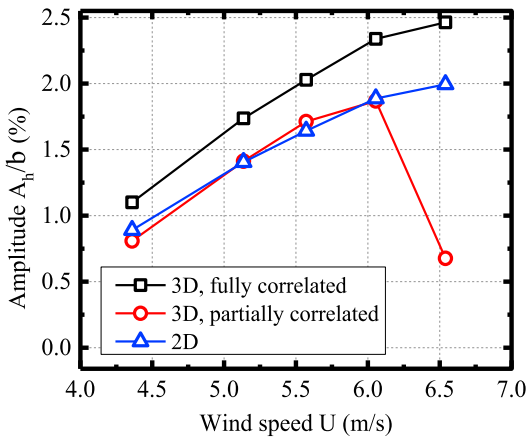
$$G_{c_{h_j h_j}}^{H_1^*} = \left[\int_0^L \int_0^L H_1^*(x_A) H_1^*(x_B) h_j^2(x_A) h_j^2(x_B) \text{cor}_{H_1^*}(x_A, x_B) dx_A dx_B \right]^{\frac{1}{2}} \quad (31)$$



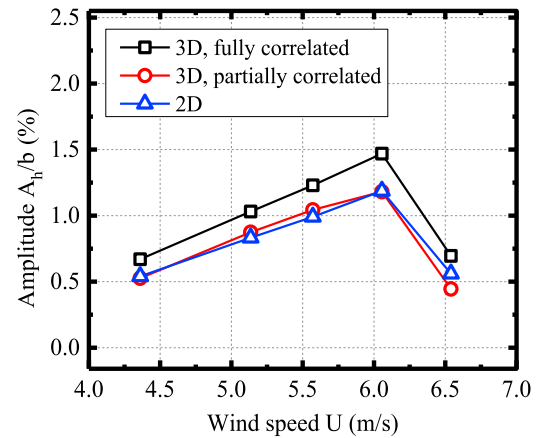
(a) $\alpha = 0$ deg



(b) $\alpha = 3$ deg



(c) $\alpha = 5$ deg



(d) $\alpha = 7$ deg

Fig. 29. Prediction of 3D vertical VIV amplitude.

and $\lambda_{H_1^*}$ represents a factor associated with spanwise correlation of vortex-induced forces. When the forces are fully correlated, $\lambda_{H_1^*} = 1$. Similar expressions for H_4^* can also be presented.

Ignoring the influence of H_4^* , the amplitude-dependent aerodynamic damping ratio with consideration of spanwise correlation can be expressed as:

$$\xi_{haj} = -\frac{1}{2} \mu \lambda_{H_1^*} G_{h_j h_j}^{H_1^*} / G_{h_j h_j} \quad (32)$$

By comparing the aerodynamic damping of 2D section, i.e., Eq. (5), it is obvious that H_1^* in 2D analysis is replaced by $\lambda_{H_1^*} G_{h_j h_j}^{H_1^*} / G_{h_j h_j}$ in this 3D analysis. The steady-state amplitude of VIV is determined at the condition that the total system damping is zero, i.e., $\xi_{hsj} + \xi_{haj} = 0$.

In the following, a suspension bridge with a main span 1700 m is used for analysis. The bridge deck with $B = 31.4$ m. Fig. 28 shows the normalized fundamental symmetric vertical mode with $G_{h_1 h_1} = 440.8$. The modal frequency is $f_{hs} = 0.1244$ Hz, $\mu = 0.0189$ and $\xi_{hs} = 0.35\%$. The decay factor $c_{H_1^*} = 5$ corresponds to an integral length of 338 m.

Fig. 29 displays the predicted steady-state vertical amplitude at the midspan center. For comparison, the 2D VIV amplitude is also presented. It is observed that the 3D estimate is significantly larger than that of 2D when the vortex-induced forces are assumed to be fully correlated.

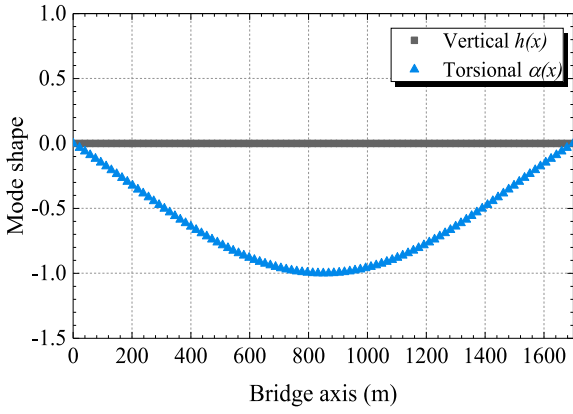


Fig. 30. Bridge torsional mode shape.

However, the 3D estimate is reduced significantly when partial correlation of forces is considered.

The fact that 3D VIV amplitude (when the aerodynamic forces are assumed to be spanwise fully correlated) is greater than 2D VIV can be explained as follows. In 3D analysis, the amplitude of vibration varies with spanwise location and is lower than the largest amplitude. The nonlinear aerodynamic damping decreases with increasing vibration amplitude. Therefore, the magnitude of total negative aerodynamic damping in 3D analysis is larger than that of 2D analysis for a given same amplitude, i.e., $G_{h_1 h_1}^{H_1^*} / G_{h_1 h_1}$, H_1^* in 2D. The VIV amplitude of higher vertical modes can also be estimated, which is not further discussed here.

6.2. Prediction of coupled VIV and flutter

The coupling of fundamental vertical and torsional modes with modal shapes $h_1(x)$ and $\alpha_2(x)$ is considered. The equations of motion in terms of generalized modal displacements are similar to those of 2D analysis with the flutter derivatives $T_p^* = H_1^*$ and H_4^* replaced by $G_{h_1 h_1}^{T_p^*} / G_{h_1 h_1}$; $T_p^* = A_2^*$ and A_3^* replaced by $G_{\alpha_2 \alpha_2}^{T_p^*} / G_{\alpha_2 \alpha_2}$; $T_p^* = H_2^*$, H_3^* , A_1^* and A_4^* replaced by $G_{h_1 \alpha_2}^{T_p^*} / G_{h_1 \alpha_2}$, and $\mu\nu$ replaced by $\mu\nu D^2$, where $D = G_{h_1 \alpha_2} / \sqrt{G_{h_1 h_1} G_{\alpha_2 \alpha_2}}$ is the similarity factor of vertical and torsional modes.

The closed-form solution (Chen and Kareem, 2006, 2007) is used to calculate the modal frequency and damping ratio of the torsional modal branch at a given wind speed and torsional amplitude. The steady-state amplitude corresponds to zero total system damping of the torsional modal branch, i.e., $\xi_2 = 0$. The same suspension bridge is used for analysis. Fig. 30 shows the normalized fundamental symmetric torsional mode with $G_{\alpha_2 \alpha_2} = 810.2$. The modal frequency is 0.3047 Hz, $\nu = 0.0532$ and $D = 0.762$. The modal damping ratios of both modes are assumed to be 0.15%.

Fig. 31 displays the steady-state amplitude of both torsional and vertical displacements as functions of wind speed and angle of attack. For coupled VIV, the 3D analysis results in larger amplitude at lower wind speed but lower amplitude at higher wind speed when aerodynamic forces are assumed to be fully correlated. Consideration of the spanwise partial correlation of forces leads to a significant reduction in predicted amplitude at angle of attack of 0° , but the reduction becomes smaller as the angle of attack increases. It is apparent that the modeling of spanwise correlation of vortex-induced forces are essential for the prediction of VIV amplitude. The 3D prediction of flutter amplitude is significantly lower than that of 2D analysis. The self-excited aerodynamic forces have strong spanwise correlation at the wind speed region apparently away from the vortex lock-in wind speed (e.g., Haan et al., 2016), thus are assumed to be fully correlated in the flutter analysis, as is generally practiced in linear flutter analysis. At zero angle of attack, there is almost no flutter observed by 3D analysis. The difference between 3D and 2D

analysis reduces at higher angle of attack.

To further understand the mechanism which is responsible for a lower amplitude in 3D flutter analysis, the contributions of different force components to the system damping are compared in both 2D and 3D analyses. The results at $U = 49.6$ m/s ($U/f_a B = 5.6$), $\alpha = 0$ and 5° are shown in Fig. 32. The 3D analysis gives lower magnitude of negative damping caused by coupling of vertical and torsional motion primarily due to mode shape similarity factor $D = 0.762$ which is less than unity. As the results, the 3D analysis leads to lower magnitude of negative total damping thus lower amplitude of self-excited flutter vibration. At $\alpha = 0^\circ$, the uncoupled motion generates positive aerodynamic damping, and the coupled flutter is the classical coupled flutter in which the coupled motion is essential for its generation. On the other hand, at $\alpha = 5^\circ$, the torsional motion itself produces negative damping and drives the system to flutter, while the coupled vertical motion further adds negative damping to the system. The flutter in this case is essentially a torsional flutter.

7. Concluding remarks

This study investigated the characteristics of vertical and torsional VIVs and flutter of a streamlined bridge deck section at different wind speed and angle of attack. The vertical VIV is featured by a pure vertical motion, while the torsional VIV and flutter exhibit coupled vertical and torsional motions with phase difference. With the increase in positive angle of attack, the wind speed ranges of vertical and torsional VIV almost remain unchanged, but the flutter onset wind speed decreases. As a result, the torsional VIV and flutter mixed together at large angles of attack with lower system damping.

This study presented a Hilbert transform-based approach for quantifying the instantaneous modal characteristics from response time history. It was revealed that modal frequency, amplitude ratio and phase difference of coupled vertical and torsional motions in torsional VIV and flutter can be treated as independent of torsional amplitude. At both VIV regions and flutter region, however, the magnitude of negative aerodynamic damping decreases with increase in vibration amplitude. The aerodynamic damping at a given wind speed and angle of attack was modeled as a polynomial function of vibration amplitude. This model is more flexible and accurate in capturing the amplitude dependence of damping as compared to currently widely used model that only involves the constant term and the term proportional to square of vibration amplitude.

This study presented an approach for extracting amplitude-dependent flutter derivatives associated with coupled vertical and torsional motions based on the modal properties of coupled motion. It was shown that the flutter derivative A_2^* , which is related to aerodynamic damping caused by torsional motion, is sensitive to torsional amplitude. It is also sensitive to angle of attack. With these nonlinear flutter derivatives, the response time history, time-varying amplitude and steady-state amplitude for a given set of structural parameters were computed. By comparing the estimations with wind tunnel data, the accuracy of proposed analysis framework was illustrated. This analysis framework can be used to examine influence of various structural parameters and vibration control strategies on VIV and flutter.

The steady-state amplitude of VIVs of 3D bridge were estimated and the significance of the consideration of spanwise variations and correlation of flutter derivatives (i.e., self-excited forces) was highlighted. The amplitude of nonlinear 3D coupled flutter was also determined, which was significantly lower than that of 2D analysis primarily due to the lower coupling effect between vertical and torsional motions in 3D analysis. At zero angle of attack, the uncoupled motion generates positive aerodynamic damping, and the coupled flutter is the classical coupled flutter in which the coupled motion is essential for its generation. At a larger angle of attack the flutter is in fact initiated from torsional motion while the coupled vertical motion further adds negative damping to the

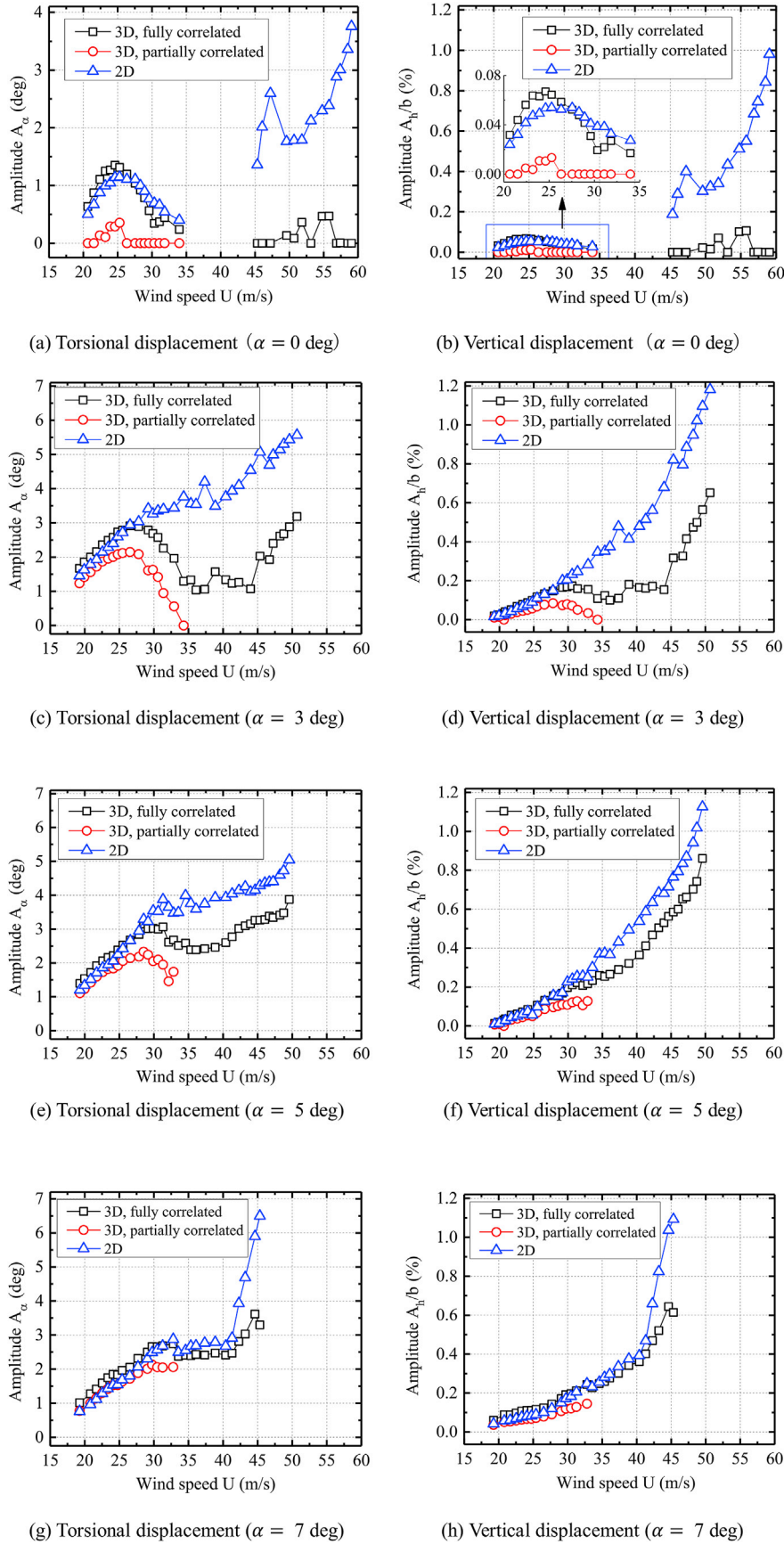


Fig. 31. Prediction of 3D coupled VIV and flutter.

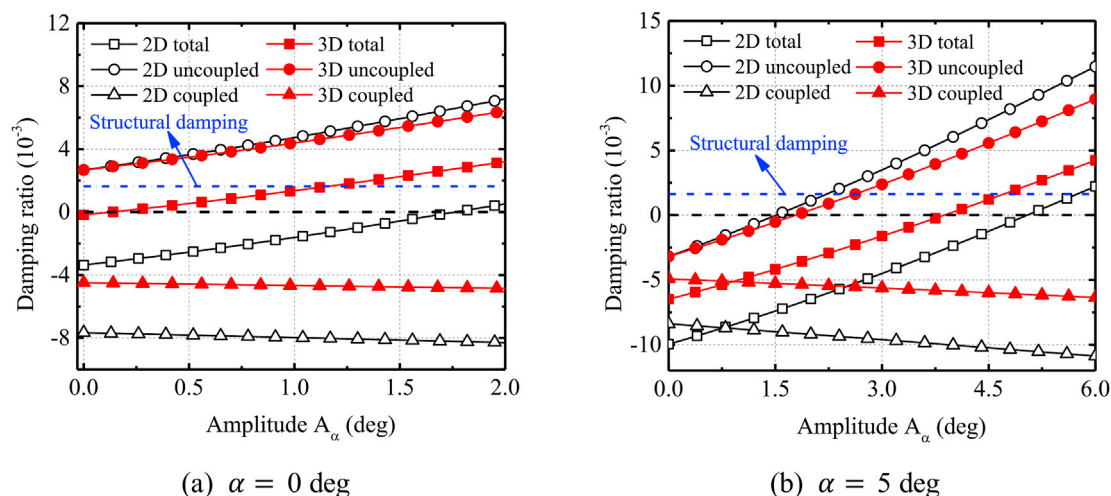


Fig. 32. Comparison of estimated damping ratios of 2D and 3D analyses (Flutter, $U = 49.6$ m/s).

system. This analysis shed new insights on the role of different aerodynamic force components and similarity factor of vertical and torsional mode shapes on the generation of negative aerodynamic damping and steady-state amplitude of VIV and flutter.

CRediT authorship contribution statement

Yunfei Wang: Formal analysis, Data curation, Writing - original draft. **Xinzhong Chen:** Supervision, Methodology, Writing - review & editing. **Yongle Li:** Formal analysis, Data curation.

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References

- Carassale, L., Wu, T., Kareem, A., 2014. Nonlinear aerodynamic and aeroelastic analysis of bridges: frequency domain approach. *J. Eng. Mech.* 140 (8), 04014051.
- Chen, X., Matsumoto, M., Kareem, A., 2000a. Time domain flutter and buffeting response analysis of bridges. *J. Eng. Mech.* 126 (1), 7–16.
- Chen, X., Matsumoto, M., Kareem, A., 2000b. Aerodynamic coupling effects on flutter and buffeting of bridges. *J. Eng. Mech.* 126 (1), 17–26.
- Chen, X., Kareem, A., 2001. Nonlinear response analysis of long-span bridges under turbulent winds. *J. Wind Eng. Ind. Aerod.* 89, 1335–1350.
- Chen, X., Kareem, A., 2002. Advanced analysis of coupled buffeting response of bridges: a complex modal decomposition approach. *Probabilist. Eng. Mech.* 17, 201–213.
- Chen, X., Kareem, A., 2003. Aeroelastic analysis of bridges: effects of turbulence and aerodynamic nonlinearities. *J. Eng. Mech.* 129 (8), 885–895.
- Chen, X., Kareem, A., 2004. Efficacy of the implied approximation in the identification of flutter derivatives. *J. Struct. Eng.* 130 (12), 2070–2074.
- Chen, X., Kareem, A., 2006. Revisiting multimode coupled bridge flutter: some new insights. *J. Eng. Mech.* 132 (10), 1115–1123.
- Chen, X., 2007. Improved understanding of bimodal coupled bridge flutter based on closed-form solutions. *J. Struct. Eng.* 133 (1), 22–31.
- Chen, X., 2013. Estimation of stochastic crosswind response of wind-excited tall buildings with nonlinear aerodynamic damping. *Eng. Struct.* 56, 766–778.
- Chen, X., 2014. Extreme value distribution and peak factor of crosswind response of flexible structures with nonlinear aeroelastic effect. *J. Struct. Eng.* 140 (12), 04014091.
- Diana, G., Rocchi, D., Argenti, T., Muggiasca, S., 2010. Aerodynamic instability of a bridge deck section model: linear and nonlinear approach to force modeling. *J. Wind Eng. Ind. Aerod.* 98 (6–7), 363–374.
- Ehsan, F., Scanlan, R.H., 1990. Vortex-induced vibrations of flexible bridges. *J. Eng. Mech.* 116 (6), 1392–1411.

- Feng, C., Chen, X., 2018. Estimation of nonstationary crosswind response of tall buildings with nonlinear aeroelastic effect. *J. Eng. Mech.* 144 (7), 04018053.
- Haan, F., Wu, T., Kareem, A., 2016. Correlation structure of pressure field and integrated forces on an oscillating prism in turbulent flows. *J. Eng. Mech.* 142 (5), 0001058.
- Hahn, S.L., 1996. *Hilbert Transforms in Signal Processing*. Artech House, Boston, MA.
- Iwan, W.D., Blevins, R.D., 1974. A model for vortex-induced oscillations of structures. *J. Appl. Mech.* 41, 168–187.
- Landl, R., 1975. A mathematical model for vortex-induced vibration of bluff bodies. *J. Sound Vib.* 42 (2), 219–234.
- Larsen, A., 1995. A generalized model for assessment of vortex-induced vibrations of flexible structures. *J. Wind Eng. Ind. Aerod.* 57, 281–294.
- Li, M.S., Li, S.P., Liao, H.L., Zeng, J.D., Wang, Q., 2016. Spanwise correlation of aerodynamic forces on oscillating rectangular cylinder. *J. Wind Eng. Ind. Aerod.* 154, 47–57.
- Matsumoto, M., Daito, Y., Yoshizumi, F., Ichikawa, Y., Yabutani, T., 1997. Torsional flutter of bluff bodies. *J. Wind Eng. Ind. Aerod.* 69–71, 871–882.
- Mashad, M., Jones, N.P., 2014. A model for vortex-induced vibration analysis of long-span bridges. *J. Wind Eng. Ind. Aerod.* 134, 96–108.
- Matsumoto, M., Nihara, Y., Kobayashi, Y., Shirato, H., Hamasaki, H., 1995. Flutter mechanism and its stabilization of bluff bodies. In: *Proc., 9th Int. Conf. On Wind Engineering*, New Delhi, India, pp. 827–838.
- Matsumoto, M., Yoshizumi, F., Yabutani, T., Abe, K., Nakajima, N., 1999. Flutter stabilization and heaving-branch flutter. *J. Wind Eng. Ind. Aerod.* 83, 289–299.
- Naprstek, J., Pospisil, S., Hracov, S., 2007. Analytical and experimental modeling of nonlinear aeroelastic effects on prismatic bodies. *J. Wind Eng. Ind. Aerod.* 95, 1315–1328.
- Noda, M., Utsunomiya, H., Nagao, F., Kanda, M., Shiraishi, N., 2003. Effects of oscillation amplitude on aerodynamic derivatives. *J. Wind Eng. Ind. Aerod.* 91 (1), 101–111.
- Scanlan, R.H., 1978. The action of flexible bridges under wind. I: flutter theory. *J. Sound Vib.* 60 (2), 187–199.
- Scanlan, R., 1997. Amplitude and turbulence effects on bridge flutter derivatives. *J. Struct. Eng.* 123 (2), 232–236.
- Sun, Y.G., Li, M.S., Liao, H.L., 2013. Nonlinear approach of vortex-induced vibration for line-like structures. *J. Wind Eng. Ind. Aerod.* 124, 1–6.
- Simiu, E., Scanlan, R.H., 1986. *Wind Effects on Structures*. John Wiley, New York, USA.
- Tamura, Y., Matsui, G., 1979. Wake-oscillator model of vortex-induced oscillation of circular cylinder. In: *Proceedings of the Fifth International Conference on Wind Engineering*, Fort Collins, Colorado, USA, pp. 1085–1094.
- Vickery, B.J., Basu, R.I., 1983. Across wind vibration of structures of circular cross section. Part I. Development of a mathematical model for two-dimensional conditions. *J. Wind Eng. Ind. Aerod.* 12, 49–73.
- Wilkinson, R.H., 1981. Fluctuating pressures on an oscillating square prism: Part II: spanwise correlation and loading. *Aeronaut. Q.* 32 (2), 111–125.
- Wu, T., Kareem, A., 2011. Modeling hysteretic nonlinear behavior of bridge aerodynamics via cellular automata nested neural network. *J. Wind Eng. Ind. Aerod.* 99 (4), 378–388.
- Wu, T., Kareem, A., 2013. Aerodynamics and aeroelasticity of cable supported bridges: identification of nonlinear features. *J. Eng. Mech.* 139 (12), 1886–1893.
- Wu, B., Chen, X.Z., Wang, Q., Liao, H.L., Dong, J.H., 2019. Characterization of vibration amplitude of nonlinear bridge flutter from section model to full bridge estimation. *J. Wind Eng. Ind. Aerod.* 197, 1040487.
- Xu, F., Ying, X., Zhang, Z., 2016. Effects of exponentially modified sinusoidal oscillation and amplitude on bridge deck flutter derivatives. *J. Bridge Eng.* 21 (5), 06016001.
- Zhu, L.D., Meng, X.L., Guo, Z.S., 2013. Nonlinear mathematical model of vortex-induced vertical force on a flat closed-box bridge deck. *J. Wind Eng. Ind. Aerod.* 122, 69–82.

- Zhu, L.D., Meng, X.L., Du, L.Q., Ding, M.C., 2017a. A simplified nonlinear model of vertical vortex-induced force on box decks for predicting stable amplitudes of vortex-induced vibrations. *Engineering* 3, 854–862.
- Zhu, Q., Chen, B.Y., Zhu, L.D., Xu, Y.L., 2017b. Investigation on characteristics and spanwise correlation of vortex-induced forces on a twin-box deck using newly-developed wind-tunnel test technique. *J. Wind Eng. Ind. Aerod.* 164, 69–81.
- Zhou, S., Hua, X.G., Chen, Z.Q., Chen, W., 2017. Experimental investigation of correlation factor for VIV amplitude of flexible bridges from an aeroelastic model and its 1:1 section model. *Eng. Struct.* 141, 263–271.
- Zhang, M., Xu, F., Ying, X., 2017. Experimental investigations on the nonlinear torsional flutter of a bridge deck. *J. Bridge Eng.* 22 (8), 04017048.
- Zhou, R., Ge, Y.J., Yang, Y.X., Du, Y.L., Zhang, L.H., 2018. Wind-induced nonlinear behaviors of twin-box girder bridges with various aerodynamic shapes. *Nonlinear Dynam.* 94, 1095–1115.